

Low back pain

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My MRI shows I have a damaged disc at L5/S1. I am disappointed that I can never play basketball, golf and go for a run ever again. A neurosurgeon says he can perform a fusion. Is there anything else I can do?

24-year-old athlete with disabling low back pain

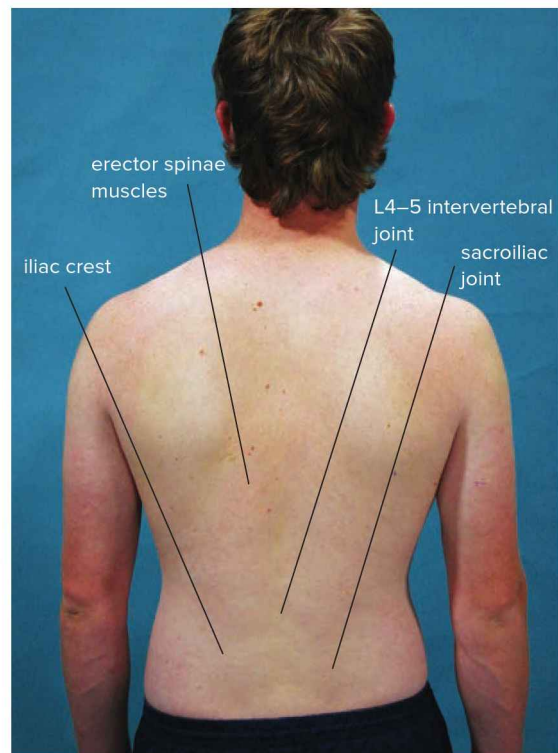
Low back pain (LBP) is the leading cause of disability in the world affecting both the general population and athletes alike. In this chapter we outline a contemporary, multidimensional understanding of LBP and detail a clinical perspective of assessing and managing the disorder.

The anatomy of the low back is shown in Figure 29.1.

EPIDEMIOLOGY

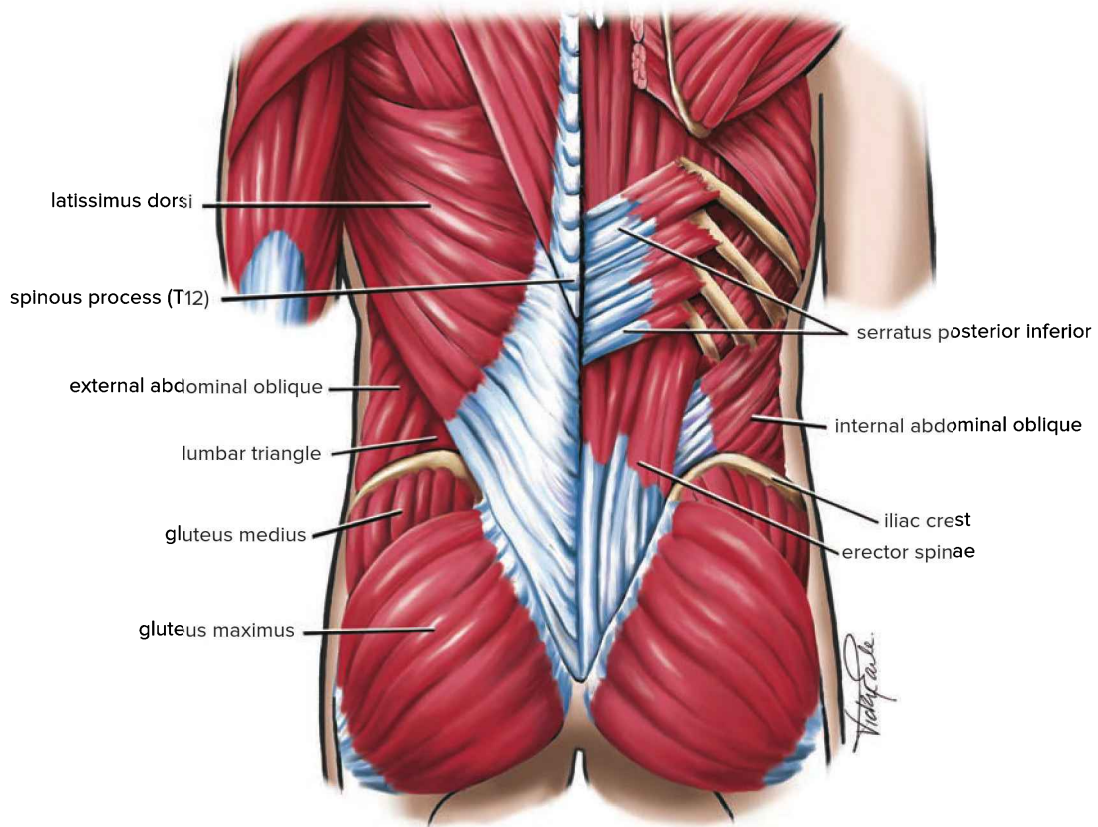
Back pain affects up to 85% of the population at some time in their lives. LBP is rarely reported before the age of 10; however, during adolescence there is a rapid increase in the reporting of LBP reaching near-adult rates (lifetime prevalence of 70%) at the age of 17 years. While the majority (70–80%) improve from an *acute* episode over a 3-month period irrespective of treatment, between 50–80% will have at least one *recurrent* episode. However, for up to 20–30%, LBP can become *persistent* and disabling, limiting sporting activity, physical functioning and impacting on a person's quality of life.¹

LBP is the most common cause of disability in those under the age of 45, and the most expensive healthcare problem in those between the ages of 20 and 50. Back problems account for a significant percentage (25% in the United States) of workers' compensation claims. LBP is also a common cause of lost time from sport. Of great concern is that in spite of enormous financial resources being spent on radiological imaging, treatments and medicating for LBP, the chronicity and disability rates are increasing.² The estimated annual cost of LBP in the United States, for example, is over US\$40 billion.

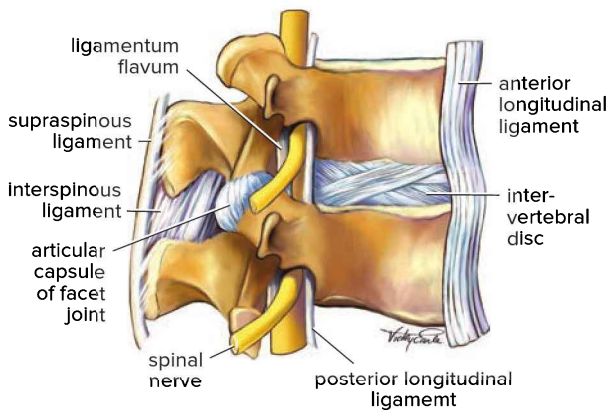


(a)

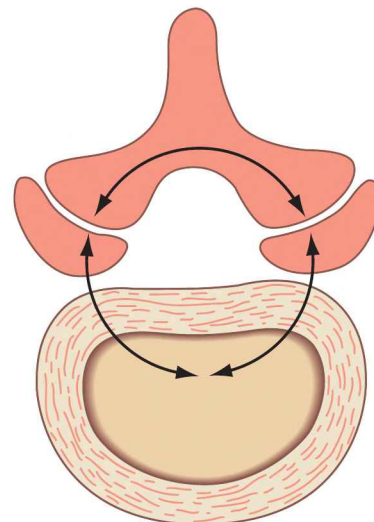
Figure 29.1 Anatomy of the low back (a) Surface anatomy



(b)

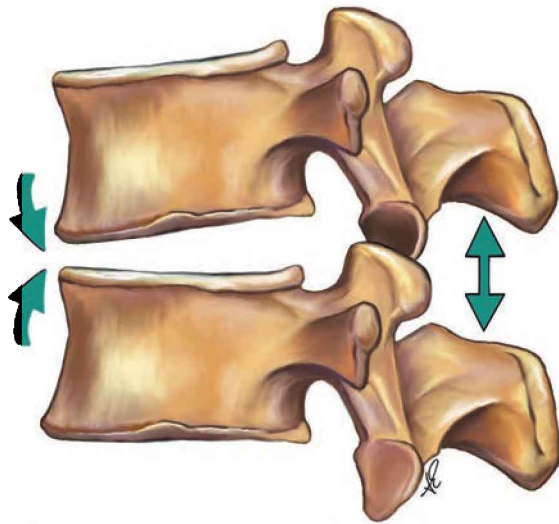


(c)

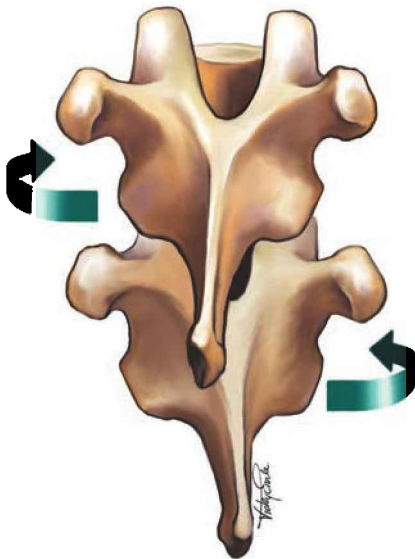


(d)

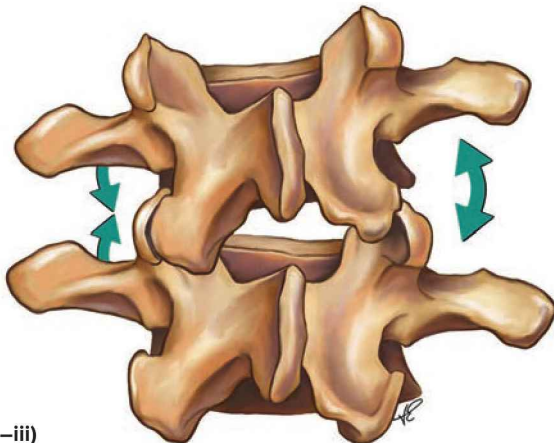
Figure 29.1 (cont.) (b) Muscles of the lower back from behind (c) The intervertebral segment (d) The three-joint complex consisting of the intervertebral disc and the two zygapophyseal (facet) joints



(e-i)



(e-ii)



(e-iii)

The multidimensional nature of low back pain

Once serious and specific pathology has been excluded, LBP in an athlete is frequently considered to be caused by an injury to the spine's structures. This is in spite of the fact that a clear mechanism of injury or even biomechanical strain often cannot be identified. Consequently, clinicians often reinforce the belief to the athlete that LBP is due to tissue damage, with treatments being directed at treating spinal structures.

In contrast, contemporary evidence supports the concept that LBP disorders in both sporting and non-sporting populations are associated with a complex combination of factors which may include pathoanatomical, physical, lifestyle, psychological, social, neurophysiological and genetic, all of which can predispose to LBP as well as maintain a vicious cycle of pain, distress and disability.^{1, 3} This may present as LBP associated with an acute injury, repeated strain or in the absence of any tissue strain. The combination of contextual risk factors for the individual will determine levels of pain experienced, disability behaviours and coping strategies, as well as the risk for pain recurrence and chronicity. A detailed understanding of the various risk factors for LBP is required in order to provide a clear framework for clinicians to both examine and direct effective treatment and management.

TRIAGE

When a person presents with LBP, the clinician must first triage him or her to rule out serious and specific pathology (Fig. 29.2, right column and left column).

Serious pathology

For 1-2% of patients, LBP may be associated with serious pathology such as malignancy, infection or inflammatory disorder. The strongest factor suggestive of malignancy is a previous history of malignancy; for infection, a previous history of intravenous drug use coupled with fever; and for inflammatory disorder, an insidious onset of non-mechanical pain linked to night pain and morning stiffness.⁴

Specific pathoanatomical diagnoses

For another 5-10% of patients with LBP, a definitive diagnosis can be made based on spinal imaging linked to careful clinical examination.

- In sporting populations, fractures related to direct trauma, such as a transverse process fracture (e.g. from a knee in the back) or compression fracture of the vertebra

Figure 29.1 (cont.) (e) Segmental motion: (i) flexion/extension (sagittal plane motion); (ii) torsion (transverse plane motion); (iii) side bending (frontal plane motion) showing compressive and torsional forces

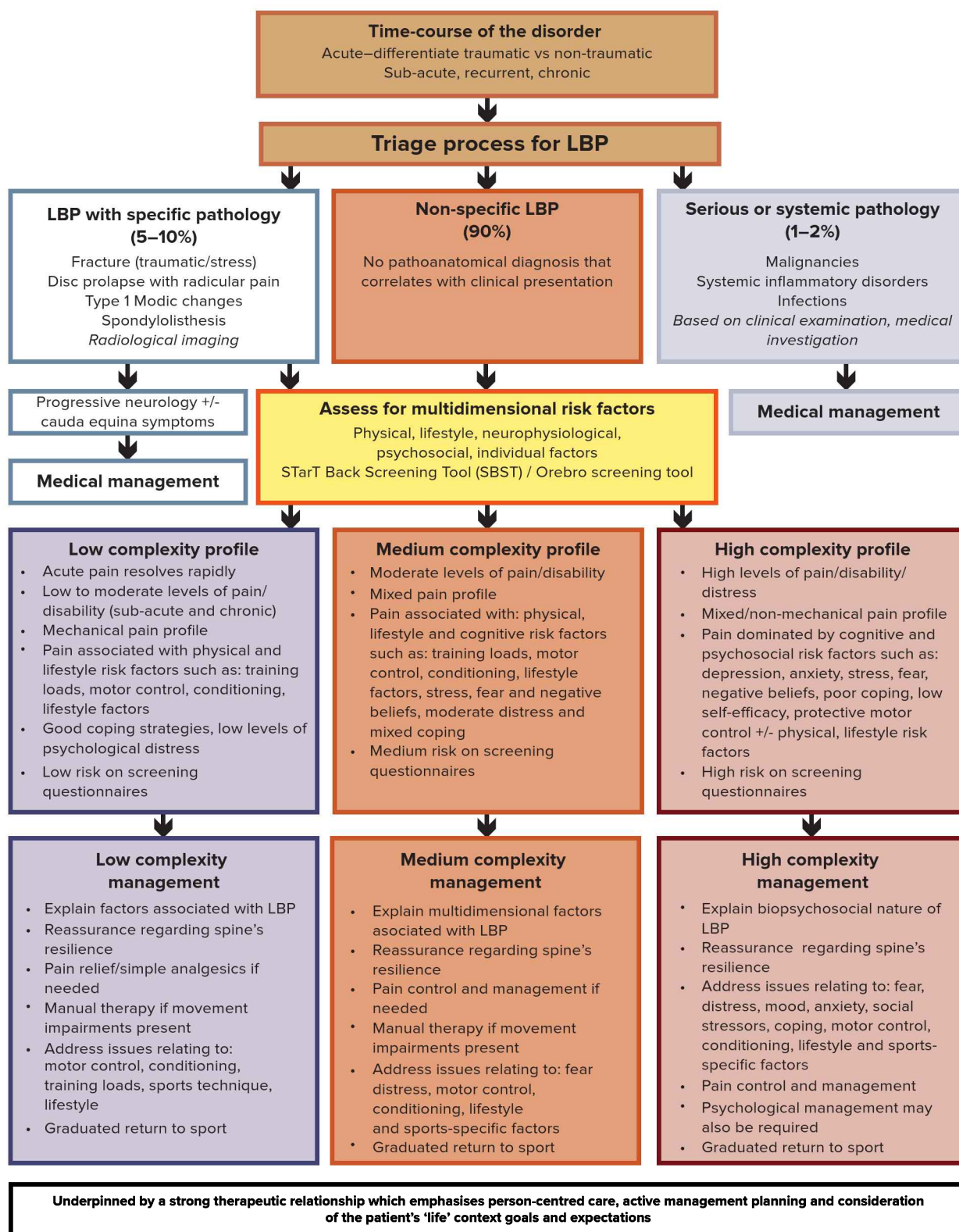


Figure 29.2 Clinical framework for the profiling and targeted management of LBP

(loading injury), can occur in the lumbar spine. Stress fractures of the pars interarticularis may also present, usually during the adolescent growth phase, in association with increased training exposures to sports that involve cyclical spinal loading such as cricket fast bowling, tennis, weightlifting, throwing sports and gymnastics. The natural history for these injuries is favourable, with conservative management involving a period of relative rest from pain-aggravating activities with a graduated return to full function.

- Spondylolysis or spondylolisthesis are present in 5% of the pain-free population and have genetic predisposition. They correlate poorly with pain and disability, but may become symptomatic in sporting populations that involve repeated spinal bending and loading. Their natural history is positive with good conservative care, unless the listhesis is high grade (>3) and/or when it is associated with neural compromise.
- While disc protrusions are present in pain-free populations, LBP associated with acute radicular pain +/- neurological deficits that correlates with imaging findings of disc prolapse, is considered a specific cause of LBP and associated leg pain. This may be linked to a history of a sudden increase in exposures to repeated spinal loading with flexion and rotation, and is considered to result in radial fissuring in people genetically vulnerable to disc prolapse. Although very painful in the acute phase, the natural history for disc prolapse is favourable and, for the majority, the prolapse resolves over time. Only in the case of progressive neurological deficit and/or cauda equina symptoms is imaging and surgical review required.
- Vertebral endplate oedema (Modic type 1 changes) is also associated with LBP and can be observed on magnetic resonance imaging (MRI) scan (T1-weighted view). It

reflects an active inflammatory process of the subchondral bone of the vertebral endplate. This may be linked to repeated biomechanical loading of an underlying degenerate disc and tends to be more prevalent in middle-aged populations. When symptomatic, this may present as disabling pain associated with loading, movement and night pain. However, the natural history for these presentations is good with appropriate conservative management.

- Spinal canal stenosis is rare in young and middle-aged athletes as the condition is related to advanced degenerative changes, but may occasionally be seen in older athletes. It is typically characterised by diffuse pain in the buttocks or legs, aggravated by standing and walking, and relieved by sitting, lying and lumbar flexion. This may be managed conservatively with exercises that reduce the lumbar lordosis and enhance hip extension mobility to reduce extension loading. Where this approach fails, surgical review may be indicated.

It must be understood that in pain-free populations, there is a high prevalence of abnormal findings on MRI scans (Table 29.1).⁵ These findings are poor predictors of future back pain and do not correlate well with levels of pain and disability.⁶ This makes a true pathoanatomical diagnosis difficult to achieve, reinforcing the need for careful clinical examination. Early MRI imaging for minor back strains results in poorer prognosis, increased instances of sick leave and a higher possibility of surgery.⁷

This highlights the potential iatrogenic influence that imaging plays in LBP and confirms that clinicians need to take great care in communicating radiology findings in a manner that is evidence based and does not create fear and anxiety, causing the patient to catastrophise their

Table 29.1 Age-specific prevalence estimates of degenerative spine imaging findings in asymptomatic patients

Imaging finding	Age						
	20	30	40	50	60	70	80
Disc degeneration	37%	52%	68%	80%	88%	93%	96%
Disc signal loss	17%	33%	54%	73%	86%	94%	97%
Disc height loss	24%	34%	45%	56%	67%	76%	84%
Disc bulge	30%	40%	50%	60%	69%	77%	84%
Disc protrusion	29%	31%	33%	36%	38%	40%	43%
Annular fissure	19%	20%	22%	23%	25%	27%	29%
Facet degeneration	4%	9%	18%	32%	50%	69%	83%
Spondylolisthesis	3%	5%	8%	14%	23%	35%	50%

Reproduced with permission from Brinjikji et al. 2015⁵

condition. Even when a specific diagnosis can be made, consideration of the multidimensional factors that can influence pain and disability (listed in the next section) is important and may need to be addressed for optimal management of the disorder.

Low back pain without a pathoanatomical diagnosis

PRACTICE PEARL

For the majority of patients who present with LBP (85–90% of patients), no definitive structural diagnosis can be made, leaving a management vacuum and a diagnosis of 'non-specific low back pain'. However, the experienced clinician can usually identify a clear pattern of tissue sensitisation linked to specific spinal structures.

FACTORS CONTRIBUTING TO LOW BACK PAIN

A major recent paradigm shift has been the recognition that non-specific back pain can be strongly associated with a variety of different modifiable physical, lifestyle, psychosocial and neurophysiological risk factors that provide the clinician with the opportunity to target management.^{1,5} We explore those in turn.

Physical factors

Extrinsic factors

Each sport carries a different risk for LBP. An increased risk of LBP is linked to specific sports associated with high volumes of sustained and cyclical loading of the spine especially when coupled with rotation and side bending. For example, sports such as cycling, field hockey and rowing are associated with increased flexion loading +/- rotation strain, whereas sports such as dance, gymnastics, tennis and fast bowling in cricket are associated with increased extension strain coupled with side bending and rotation. These sport-specific factors interplay closely with training loads, volumes and technical aspects of the sport, which are closely related to the reporting of LBP.⁸ Considering these extrinsic factors in conjunction with the coach and sport demands, and in line with the intrinsic factors of the individual, is important for targeted management.

Intrinsic factors

There is evidence that non-neutral spinal postures in standing (sway, hyperlordotic and flat back) are associated with an increased risk of LBP.⁹ These postures are linked to changes in spinal loading as well as altered motor control patterns of the trunk and the hip region.¹⁰ There

is also evidence that these habitual postures influence movement patterns and may represent a 'signature' upon which more complex motor patterns develop.¹¹ These postural patterns, when interacting with sport-specific related extrinsic factors, may result in end-range tissue strain sensitising the spinal structures. This can result in directional patterns of pain sensitisation (flexion/extension/lateral [side bending] or multidimensional) in the presence or absence of tissue injury.¹²

In the acute phase of an LBP disorder following a traumatic injury and/or when associated with pathology, muscle guarding and movement impairments may be protective and adaptive. However, in the absence of tissue injury or when pain persists past tissue healing time, 'protective muscle guarding' may become unhelpful, provocative and maladaptive. This may reflect a response to a person's perception of underlying sense of threat, fear or vulnerability in the absence of ongoing pathology or tissue injury.^{12,13}

In this case, increased protective trunk muscle co-contraction may increase spinal stiffness and compressive loading, which may provoke pain-sensitive structures. This is similar to the 'limp' that occurs when a person sprains their ankle and that then persists beyond the natural tissue healing time, predisposing the person to recurrent and/or persistent pain.

Persistent motor control deficits (as described below) can also potentially leave a person vulnerable to increased tissue loading. These patterns are not stereotypical and are commonly associated with directional patterns of sensitisation with spinal movement and loading.^{12,13}

There is also evidence that people with LBP present with proprioceptive deficits specific to these patterns,¹⁴ as well as changes in the cortical mapping of the motor cortex,¹³ which may act to reinforce the maintenance of provocative patterns of movement. This highlights the need for an individualised mind-body approach to rehabilitation.

Deconditioning and muscle weakness in one body region, secondary to habitual postures and movement patterns, avoidance or sedentary behaviours, can also alter patterns of motor control in a proximal region. For example, weakness of the gluteal or lower limb muscles has been shown to increase lateral trunk shift and increased levels of trunk muscle co-contraction with single-leg loading resulting in increased spinal loading.¹⁵ This may be linked to pain associated with sports involving limb loading. On the other hand, deficits in back muscle strength and endurance may render a person vulnerable to flexion loading strain in a sport such as rowing or hockey.^{14,16}

Lifestyle factors

Sedentary behaviour, sudden increases in activity levels, abdominal obesity, sleep deficits and disruption, excessive alcohol and smoking are all known to be risk factors

that can act both peripherally and centrally to sensitise spinal structures.¹⁷⁻¹⁹ These may be important targets for management.

Psychosocial factors

Cognitive factors

Negative beliefs, as well as fear of pain, movement and activity, are strong predictors of disability levels.²⁰ Healthcare professionals provide a critical role in transferring beliefs with regard to back pain to their patients. Language such as 'your back is unstable' may be interpreted as 'my back is damaged and it's dangerous to move'. A comment that states a 'lack of core stability' may lead the patient to believe that 'my back is weak and vulnerable and I need to be vigilant to protect when I move'. Words such as 'disc degeneration' and 'wear and tear' may also have long-lasting negative effects.^{21, 22}

These may contribute to: fear of pain, movement and sporting activity; pain hypervigilance; and catastrophising linked to the belief that pain equals harm. Low levels of confidence and self-efficacy are also related to higher disability levels and this is often related to poorer coping strategies.²³ In some athletes, this may manifest as 'avoidance' of movement and/or sporting activity and dependence on passive treatments, while for other athletes 'endurance' coping strategies may present where they overexert themselves with pain-provoking activities in spite of significant pain exacerbation.²⁴ This can lead to an activity 'boom' (i.e. overactivity), followed by 'bust' (i.e. inactivity) pattern.

The responsibility of the clinician is to use language and management strategies that build confidence and empower the patient to develop effective self-management strategies while reducing reliance on passive therapies.²¹ It is known that negative back pain beliefs and fear are also associated with avoidance behaviours and protective muscle guarding highlighting the close body-mind relationship with LBP.²⁵ Inquiring and screening for an athlete's pain beliefs, levels of fear and thoughts about pain are essential for effective management.^{21, 26}

Emotional factors

Emotional factors such as stress, anxiety, depression, anger and frustration can act to reinforce maladaptive behaviours, further enhancing the pain experience and disability levels.³ They can also influence pain processing, both centrally and peripherally, through dysregulation of the hypothalamic-pituitary-adrenal axis, altered immune and neuro-endocrine function, and cortical changes. They are commonly associated with LBP in sporting populations²⁷ and require screening and enquiry.²⁶ Screening for comorbid mental health disorders (anxiety disorder, obsessive compulsive disorder, panic disorder and depression) is important, especially when an athlete is disabled and distressed.

Social stressors

Work- and sports-related stress, expectations of the coach, media, family stress and cultural factors can all have an influence on pain beliefs, coping and vulnerability and need to be considered in the context of the athlete.³ Significant life stress events are predictive of LBP disorders and should be screened for when the pain experience is disproportionate to the pain history (i.e. disproportionate pain responses to mechanical loading) and when pain develops in the absence of a clear history of tissue loading.

Neurophysiological factors

LBP is associated with complex changes in both the peripheral and central nervous system. There is growing evidence that persistent LBP is also associated with changes in the neuro-immune and neuro-endocrine systems, sensorimotor cortex, and associated with a loss of endogenous pain inhibition related to genetic/environmental interactions.³ These factors are thought to contribute to the various sensorimotor changes observed by clinicians in a clinical setting, ranging from anatomically specific to more widespread pain disorders.²⁸

Proportionate pain responses to mechanical loading

In sporting populations, LBP disorders are often localised to an anatomical region and present with proportionate pain responses relative to specific patterns of biomechanical loading (i.e. pain is clearly linked to directional patterns of spinal loading and movement).²⁸ There is growing evidence that cyclical spinal loading that involves either end-range or combined spinal movements (sagittal with side bending and/or rotation) with compressive loading can result in pain sensitisation of spinal structures without demonstrable 'tissue damage'.

In these cases, any of the nociceptive structures of the lumbopelvic region may be associated with LBP. These structures may include the vertebral endplate, annulus fibrosis of the intervertebral disc which are the main load-bearing structures, the laminae, zygapophyseal joints and surrounding ligamentous structures which provide resistance to end-range loading and torsion, the dura mater and nerve structures, as well as muscles and their fascia. All of these structures are richly innervated and may become sensitised by different combinations of the multidimensional risk factors reported earlier.

In the case of referred or 'neurogenic' pain where leg pain presents with signs of neural tissue sensitivity (positive single-leg raise [SLR] and slump tests), this may be associated with sensitisation of the nerve root epineurium in the absence of nerve root compression.²⁹ The potential lumbar spine nociception generators are listed in Table 29.2. By using skillful palpation, the clinician may be able to localise pain to discrete spinal levels associated

Table 29.2 Lumbar spine pain generators

- Nucleus pulposus
- Annulus fibrosus
- Facet joints
- Ligaments
- Muscles
- Nerve
- Synovium

with localised muscle tension and tenderness (trigger points); however, the ability to identify the sensitised structure/s with confidence (i.e. disc versus facet joint) based on clinical examination is limited.

Disproportionate pain responses to mechanical loading

Inflammatory pain is usually a sequel to tissue injury or trauma. It is characterised by constant pain at rest that may be provoked with movement or loading. This pain should settle with expected healing time and may respond to non-steroidal anti-inflammatory drugs (NSAIDs). Inflammatory pain may also be associated with a repetitive tissue strain mechanism (i.e. after an hour on a rowing ergometer or bowling 10 overs in cricket).

For others with LBP, their symptoms may be more widespread, variable and demonstrate disproportionate pain responses relative to biomechanical loading (i.e. pain that is provoked with minimal biomechanical stress or aggravated by levels of stress, anxiety, low mood, disrupted sleep and/or fatigue rather than biomechanical factors). In these cases, pain is often more bothersome, distressing and disabling; and the underlying pain mechanisms are commonly linked to central pain processes (central sensitisation) and widespread tissue hypersensitivity rather than clear nociceptive processes.

This may present clinically as pain provoked by minor mechanical provocation in all directions of spinal movement and/or loading, and associated with widespread tissue hyperalgesia and multiple trigger points in surrounding muscles. On the other hand, it may present as a person who is distressed, anxious and hypervigilant, but with an absence of clear clinical signs.

The identification of widespread tissue sensitivity to palpation, cold and/or movement may be clinical indicators of central sensitisation. Localisation of pain to discrete spinal levels is usually difficult in these cases. These presentations are also more likely to present with comorbid anxiety, depression, heightened levels of distress as well as non-specific health complaints (i.e. fatigue, nausea, headaches, stomach pain, sleep disturbance and so on).

Treating the sensitised tissue structures (with injections or manual techniques) may result in short-term relief, be

unsuccessful or indeed exacerbate the pain disorder, resulting in more distress. In contrast, care that targets the underlying mechanisms of the disorder (i.e. sleep, mood and stress-related factors) is required and likely to be more effective.

Many with LBP may present with a mixed pain picture where consideration of all multidimensional factors is required.

Individual considerations

There is growing evidence to support the idea that genetic variations have a powerful influence on pain processing as well as tissue structure (i.e. what you see on an MRI scan).³ These factors interact with biopsychosocial factors in the presentation of LBP. This can explain how two people with similar MRI findings can have very different pain experience and disability levels.

The presence of health (irritable bowel, fatigue, restlessness) and pain (headaches, migraines) comorbidities are important as they may reflect the level of the allostatic load on the athlete (physiologic consequences of adapting to repeated or chronic stress).³⁰ It has been identified that these factors are predictive of episodes of LBP which may be mediated via neuro-immune processes.

The athlete's goals, expectations and readiness for change are also known to be important considerations in the assessment, management and prognosis of people with LBP and need to be considered.^{31, 32}

CLINICAL APPROACH

A clinical reasoning process, based on the patient's 'story', screening questionnaires²⁶ and clinical examination, determines the relative weighting, dominance and relevance of the variety of factors with regard to the individual's disorder (Fig. 29.2). This allows the clinician to prioritise and target treatment and management.

History

The aim of the interview is to hear the story of LBP from the patient's own perspective. This involves the pain history, gaining a clear understanding of the mechanism of injury or biomechanical aggravation if present. Where LBP is of insidious onset, it is essential to understand changes in training load, contextual stressors and lifestyle factors.

Determination of the location of the pain, any radiation to the buttocks or legs, the aggravating and relieving factors (postures, movements and activities), and the presence of any associated features including sensory and motor symptoms is important. Any previous history of back pain and response to treatment in the past is noted. Understanding directional patterns of pain provocation and relief (such as flexion, extension or combinations of movement and/or loading) and how easily the pain is aggravated is important in gaining an insight into patterns

of tissue sensitivity, and for directing the physical examination as well as the type and intensity of treatment.

Where LBP is distressing and/or disabling, screening for psychosocial factors including levels of stress, anxiety, fear, focus on pain and mood are critical as well as levels and patterns of sleep. Understanding the person's beliefs about their pain, fear of pain and/or movement, coping strategies, as well as any patterns of avoidance or persistence provides insight into their levels of self-efficacy.

Understanding personally relevant goals and expectations for treatment as well as their current levels of training, activities of daily living and future sporting commitments is also important. Screening for general health, levels of energy/fatigue, comorbid pain and health complaints is necessary.

Review of radiology, where present, allows for consideration of imaging relative to the clinical presentation, as well as an opportunity to reassure the patient

regarding their spine's structural integrity if no specific pathology is present.

Screening for potentially serious symptoms include:

- cauda equina symptoms (e.g. bladder or bowel dysfunction)
- spinal cord symptoms (e.g. difficulty walking, tripping over objects)
- sensory symptoms (e.g. pins and needles, paraesthesia)
- motor symptoms (e.g. muscle weakness)
- systemic symptoms (e.g. weight loss, malaise)
- severe night pain.

The use of standardised self-reporting measures for pain (Numeric Pain Rating Scale (NPRS), body chart), disability (patient-specific functional scale, Oswestry Disability Index or Roland Morris Disability Questionnaire), psychosocial distress (STarT Back Screening Tool, Orebro multidimensional screening tool) are also recommended as screening tools to identify the complexity profile of the athlete and as a way of quantifying the effects of treatment (Table 29.3). Having the

Table 29.3 Patient-reported outcome measures (PROMs) for LBP

Roland Morris Disability Questionnaire

- Self-administered questionnaire that evaluates pain and function in LBP³³
- 24 items
- Score range: 0–24
- Test-retest reliability: 0.91³⁴ (same day), 0.93³⁵ (1–14 days) and 0.86³⁶ (3–6 weeks)
- Construct validity: SF-36 physical functioning 0.85,³⁷ SF-36 role – physical 0.70;³⁷ SF-12 physical component summary scale 0.80³⁷
- Minimal detectable change (MDC)95%: 8.6–9.5³⁸
- Minimal clinically important difference (MCID): 4³³
- Available online: www.rmdq.org

Orebro Musculoskeletal Disability Questionnaire

- Self-administered questionnaire to assist in early identification of yellow flags and if an individual is at risk of developing work disability due to pain^{39–41}
- 25 items
- Score range: 0–210
- Test-retest reliability: 0.80³⁹
- MDC95%: NA
- MCID: NA
- Available online: www.worksafe.vic.gov.au/health-professionals

Orebro Musculoskeletal Disability Questionnaire

- Self-administered questionnaire to distinguish subgroups of patients with LBP based on the presence of physical and modifiable psychosocial prognostic variables⁴²
- 9 items (4 regarding physical symptoms and 5 regarding psychosocial contributors)⁴²
- Score range: 0–9
- Test-retest reliability: overall score 0.73–0.79⁴³/psychosocial subscale 0.69–0.76⁴³
- MDC95%: NA
- MCID: NA
- Available online: www.keele.ac.uk/sbst/startbacktool

continued

Table 29.3 *Cont.***Disability Anxiety Stress Scale (DASS)**

- Self-administered questionnaire evaluating anxiety, depression and tension/stress
- 42 items (14 depression, 14 anxiety and 14 tension/stress)
- Score range: 0–126
- Test-retest reliability: depression 0.71, anxiety 0.79 and stress 0.81⁴⁴
- MDC95%: NA
- MCID: NA
- Available online: www2.psy.unsw.edu.au/dass/down.htm

Tampa Scale for Kinesiophobia

- Self-administered questionnaire evaluating fear of movement and re-injury⁴⁵
- 17 items
- Score range: 17–68
- Test-retest reliability: 0.93 (CLBP),⁴⁵ 0.78 (acute LBP)⁴⁶
- Construct validity: pain catastrophising scale 0.51/FABQ physical activity 0.53
- MDC95%: 9.2⁴⁷
- MCID: NA
- Available online: www.worksafe.vic.gov.au/health-professionals

Tampa Scale for Kinesiophobia

- Self-administered questionnaire to determine the presence of neuropathic pain in adults with LBP⁴⁸
- 9 items (7 symptoms, 1 pain course pattern and 1 pain radiation)⁴⁸
- Score range: 0–38
- Test-retest reliability: 0.93⁴⁹
- Predictive validity: Sn 0.85⁴⁸/Sp 0.80⁴⁸
- MDC95%: NA
- MCID: NA
- Available online: www.pfizerpatientreportedoutcomes.com/order-measures

NA = Not available

patient fill in at least the body chart, as well as a disability and a psychosocial screening questionnaire prior to examination is considered best practice.

Physical examination

Examination of the patient with LBP may include the following test elements.

- Assessment of functional postures, movements and sports-specific activities nominated during the interview as pain provoking and relieving. This is first carried out to observe for patterns of pain provocation, motor control strategies, guarded movement, protective behaviours, patterns of avoidance and levels of confidence.
- Assessment of the pattern, timing and range of spinal movement in relation to the whole kinetic chain. This is important in order to identify the adopted movement strategies.
- Repeated and/or sustained movements can be tested to determine patterns of pain provocation and easing (in the case of central back pain), as well as pain centralisation (where symptoms move peripheral to central, seen as a positive clinical finding) and peripheralisation (where symptoms move peripherally, seen as a negative clinical finding) in the case of leg pain. Finding a patient's directional preference or centralising the patient's pain during the physical examination can help guide management.⁵⁰
- Combined movement testing can be useful to establish patterns of pain provocation related to combined spinal movement and loading. For example, in a tennis player with LBP related to the serve, combined movement testing of backward bending combined with side bending and rotation may reproduce symptoms.
- Palpation of the back and trunk muscles is to determine levels of muscle tension during spinal movements. For example, where the back muscles are activated during

backward bending or the abdominal wall activates during forward bending, this is seen as an abnormal muscle bracing motor control strategy.

- Functional tests can be applied during these provocative functional tasks to determine whether altering the spinal posture, the movement pattern or level of muscle tension increases or reduces the pain experience during functional tasks. For example, where pain is related to lumbar spine flexion during lifting/loading tasks, enhancing the anterior pelvic tilt with relaxation of the thoracic spine can be tested during lifting to determine whether this reduces the pain experience. In contrast, where pain is related to lumbar spine extension during lifting/loading/backward bending tasks, then enhancing posterior pelvic tilt to reduce lordosis can be tested. Where pain is related to loading in neutral spine positions, relaxation of the abdominal wall with diaphragm breathing in conjunction with a relaxed thoracic posture can be used. Where pain is provoked by frontal or rotation plane activities then modify these activities by addressing control within these planes. These functional tests can provide an insight as to whether the identified motor control strategies (i.e. protective muscle guarding) protective (adaptive) or provocative (maladaptive).
- Assessment of spine 'stiffness' and patterns of tissue sensitivity in muscles and joints with palpation examination.
- Screening for lower limb (squat and lunge) strength and endurance is conducted where pain is associated with sustained and/or repeated spinal loading (i.e. running, jumping, limb loading).
- Screening for back muscle strength and endurance (squat hold or Sorensen test) is conducted where there is a pain associated with a clear loss of control into lumbar flexion.
- Testing for neurological deficits in the presence of radicular pain and signs of nerve compression.

These examination goals can be met by following this clinical routine.

1. Observation movement patterns in sitting, lying, standing, bending
2. Active movements
 - a. flexion (Fig. 29.3a)
 - b. extension (Fig. 29.3b)
 - c. lateral flexion (Fig. 29.3c)
 - d. combined movements—quadrant position (Fig. 29.3d)
 - e. single-leg extension (Fig. 29.3e)
3. Passive movements
 - a. overpressure may be applied at the end of range of active movements

b. muscle length (e.g. psoas, hamstring, gluteals)

c. hip quadrant

4. Palpation

- a. spinous processes (Fig. 29.3f)
- b. over the zygapophyseal joints
- c. inferior sulcus sacroiliac joint (Fig. 29.3g)
- d. over the iliolumbar ligament
- e. paraspinal muscles (Fig. 29.3h)
- f. quadratus lumborum (Fig. 29.3i)
- g. gluteal muscles

5. Assessment of strength

- a. lower limb (e.g. double leg squat, single-leg squat; Fig. 29.3j, lunge)
- b. Sorensen (or Biering-Sorensen) test (Fig. 29.3k)

6. Special tests

- a. straight leg raise/slump test (Chapter 15)
- b. prone knee bend/femoral slump
- c. sacroiliac joint stress tests and active straight leg raise (Chapter 30)
- d. neurological examination

The clinical utility of various clinical assessment findings are outlined in Table 29.4.



(a)

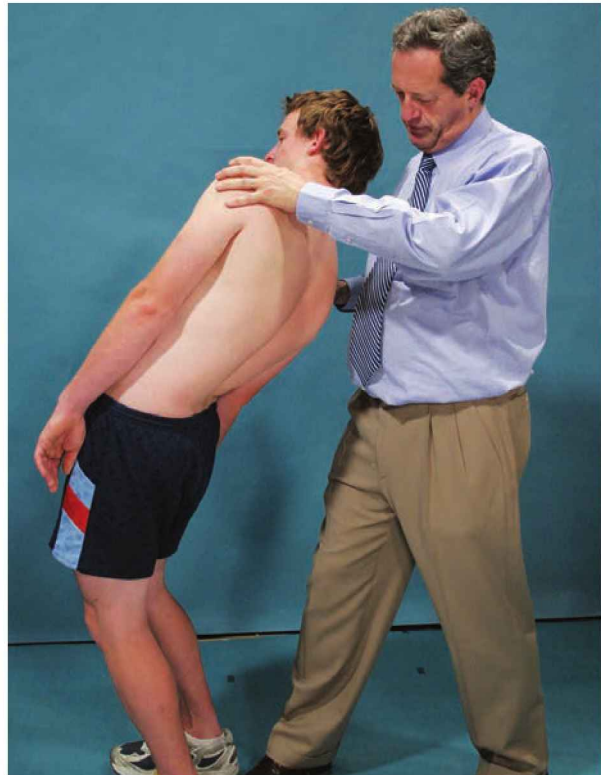
Figure 29.3 (a) Active movement—flexion. Look at symmetry of movement on both sides of the back, range of movement and, if restricted, whether it is due to pain or stiffness



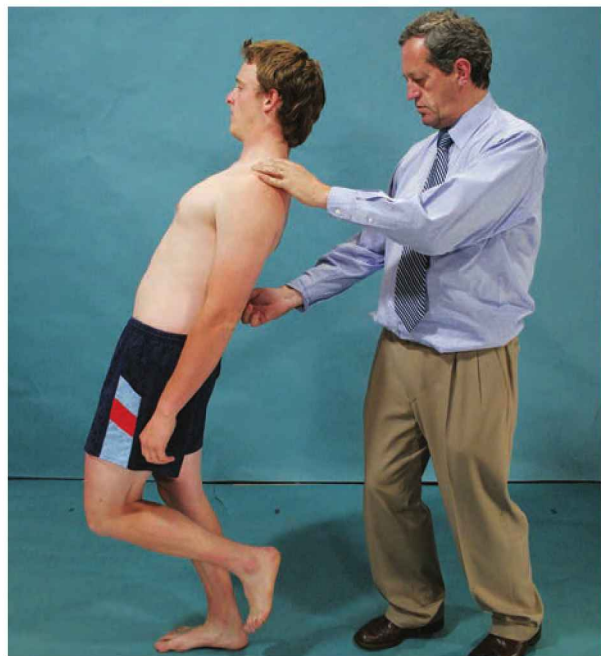
(b)



(c)



(d)



(e)

Figure 29.3 (cont.) (b) Active movement—extension. Assess degree of lumbar extension and any symptoms provoked. Patient should maintain pelvis in neutral position (c) Active movement—lateral flexion (d) Active movement—combined movement (quadrant position—extension, lateral flexion, rotation) (e) Single-leg extension



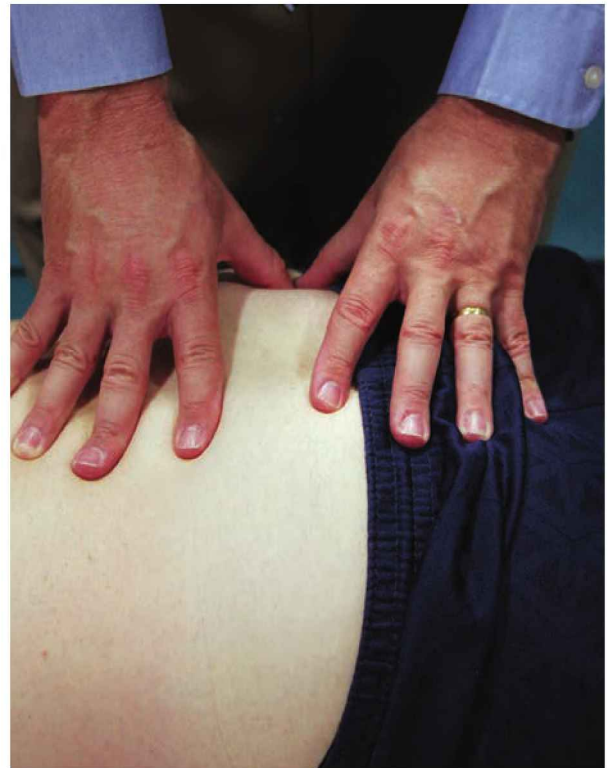
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(h)

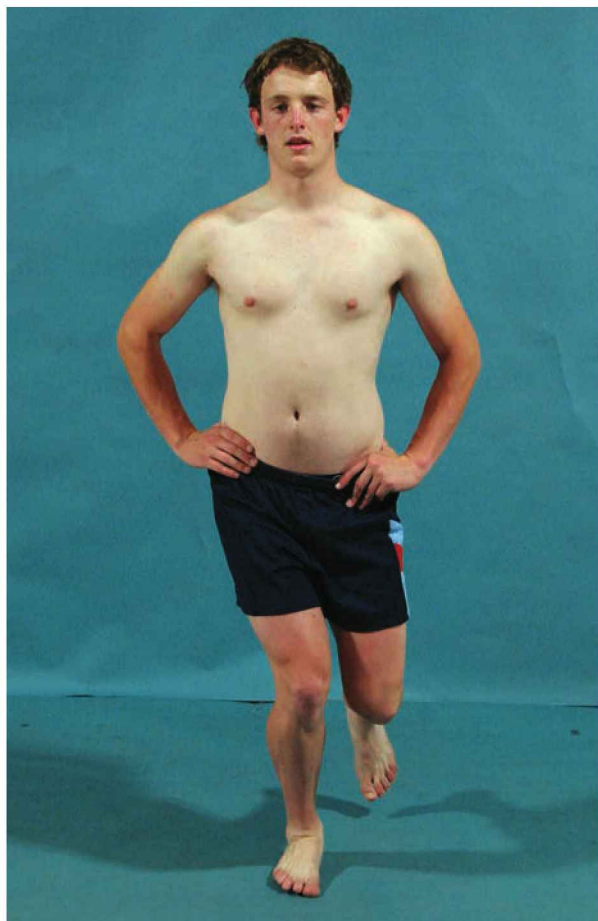


(g)



(i)

Figure 29.3 (cont.) (f) Palpation—intervertebral joints. Palpate over spinous processes, apophyseal joints and transverse processes. Assess degree of tenderness and amount of muscle guarding (g) Palpation—sacroiliac region. Palpate over the sacroiliac joints and iliolumbar ligaments (h) Palpation—muscles and fascia. Palpate paraspinal and gluteal muscles for generalised increase in muscle tone and focal areas of tissue tension, including active trigger points (i) Palpation—quadratus lumborum between the iliac crest and the rib cage



(j)

Investigations

In the management of most cases of LBP, investigations are not required. However, there are certain clinical indications for further investigation. Radiography should be performed if traumatic fracture or spondylolisthesis is suspected. Radioisotopic bone scan may be helpful in cases of suspected stress fracture of the pars interarticularis.

When specific pathology is suspected or LBP is distressing, disabling and progressive, and further diagnostic imaging is needed, MRI is often the imaging modality of choice. MRI can evaluate for a variety of possible sources of nociception including lumbar disc herniation, nerve root compression, spinal stenosis and bony stress reactions. MRI can also be used to image the internal structure of the discs. Degenerated discs that have lost fluid have a characteristic appearance on MRI (Fig. 29.4). MRI may confirm the presence of bone oedema of the vertebral endplate. The clinician must be wary, however, of placing too much emphasis on this investigation, as abnormalities shown on MRI may not necessarily be responsible for all or any of the patient's pain—they commonly reflect normal age-related changes.⁵⁷ MRI studies with and without contrast can be beneficial in patients with prior history of lumbar spine surgery and to help identify malignancy or metastatic disease.^{58, 59}

Unlike MRI, computed tomography (CT) scanning is unable to provide further information on the internal structure of the intervertebral disc, yet it can be used in cases of suspected nerve root compression where MRI is either contraindicated or unavailable. The presence and



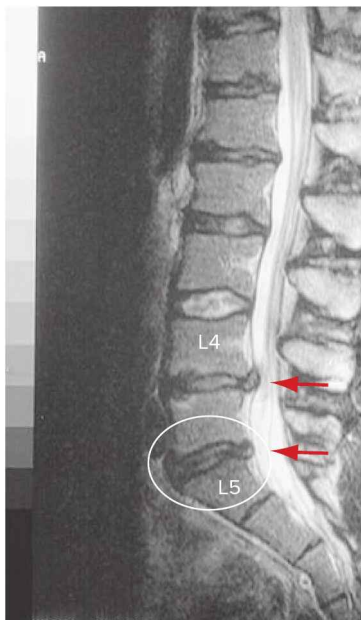
(k)

Figure 29.3 (cont.) (j) Single-leg squat functional test—double-/single-leg squat. If the patient's pain has not already been reproduced, functional tests such as squat, lunge, hop, step-up, step-down or eccentric drop squat should be performed (k) Sorensen test—patient maintains the horizontal position for as long as possible or for a set time (e.g. 240 seconds)

Table 29.4 Diagnostic accuracy of clinical tests (history, physical exam) in patients with low back pain

Test	What is a positive finding?	Sensitivity	Specificity	LR+	LR–
Lumbar disc herniation with radiculopathy					
Sensory deficits ⁵¹	Altered or reduced sensation on side of suspected radiculopathy	40	59	1.1	0.93
Motor paresis ⁵¹	Altered or reduced strength on side of suspected radiculopathy	22	79	1.1	0.96
Reflex deficits ⁵¹	Altered or reduced reflexes on side of suspected radiculopathy	29	78	1.3	0.87
Passive straight leg raise ⁵²	Pain below the knee at any angle	92	28	1.3*	0.28*
Crossed straight leg raise ⁵²	Pain below the knee at any angle	28	90	2.8*	0.8*
Slump test ⁵³	Reproduction of radicular pain	84	83	4.9	0.19
Ankylosing spondylitis					
Measurement of chest expansion ⁵⁴	<2.5 cm change from maximum expiration to maximum inspiration	91	99	91	0.09
Spondylolysis					
Single-leg hyperextension test ⁵⁵	Pain in lumbar spine	50–73	17–46	0.7–1	1.0–1.5
Spondylolisthesis					
Lumbar spinous process palpation ⁵⁵	Presence of a lumbar spinous process step deformity	60	87	4.7	0.5
Zygapophyseal joint pain					
Extension rotation test ⁵⁶	Pain with either active motion or passive overpressure of movement	100	22	1.3	0.00

LR+ = Positive likelihood ratio; LR– = Negative likelihood ratio; NA = not available; * = estimated from available research data
 See Chapter 14 for a guide to interpreting test sensitivity, specificity and likelihood ratios



acuity of a pars interarticularis defect can be determined with reverse gantry CT scanning.

MANAGEMENT OF SPECIFIC LBP DISORDERS

Stress fracture of the pars interarticularis/lumbar spondylolysis

Lumbar spondylolysis is a term used to describe radiologically confirmed lesions in the posterior vertebral arch, in the region of the pars interarticularis.^{60–64} The term includes a spectrum of bone stress injuries (stress reactions and stress fractures) and defects (non-united fractures). Traditionally spondylolysis was considered

Figure 29.4 Sagittal T2-weighted image of the lumbar spine showing decreased fluid (dark signal) in the L4/5 and L5/S1 intervertebral discs. Also noted are small disc herniations (arrows) at both levels. The clinical significance of these findings is debated (see below)

an injury of the pars interarticularis, but with the advancement of imaging techniques it is apparent that it commonly involves the adjoining pedicle and lamina.^{65, 66}

The incidence of spondylolysis in the general population is 4-9% and varies with race and sex.^{60, 67-70} In the athletic population, the incidence of spondylolysis is much higher, affecting predominantly adolescents involved in sporting activities that require repetitive spinal loading.^{71, 72} It is particularly common in tennis players, platform or springboard divers, gymnasts, soccer players, volleyball players, track and field athletes and ballet dancers.⁷³⁻⁷⁹ Cricket fast bowling, which involves a combination of repetitive trunk rotation, extension and lateral flexion during the delivery stride, has one of the highest incidence of spondylolysis of any sport.⁸⁰⁻⁸⁴ The biomechanics of cricket fast bowling are described in Chapter 9.

The reports of spondylolysis in the athletic populations can sometimes be misleading because they often include symptomatic and asymptomatic spondylolysis together, and there is typically no differentiation between active bone stress injuries (stress reactions and stress fractures) and symptomatic chronic non-united fractures or defects.

PRACTICE PEARL

When considering spondylolysis it is important to have a clear understanding of the terminology, pathology, clinical presentation and radiology because the management of spondylolysis will vary depending on the morphology of the spinal abnormality.⁶²

Terminology

The terminology used to subcategorise spondylolysis is not universal, but there are basically two main types of lesions that are distinctly different in regards to pathology, clinical presentation and radiological features.^{60, 62}

1. Active bone stress lesions. These can be subcategorised to reflect the progression along the bone stress continuum:
 - a. pre-spondylolytic bone stress
 - b. stress reaction
 - c. stress fracture.
2. Non-united stress fractures. These are often referred to as chronic defects.

The broad spectrum of spondylolytic pathology is reflected in vastly different clinical features ranging from asymptomatic to debilitating pain.^{85, 86} The clinical presentation is also typically varied and complex; and best appreciated by the radiological characteristics that traverse the bone stress continuum (Table 29.5).

The earliest stage in the bone stress continuum is *pre-spondylolytic bone stress* that is characterised by the absence of symptoms, but a radiological appearance of bone stress on imaging such as MRI or nuclear medicine but with no fracture line (Table 29.5). The absence of symptoms means that it is a condition typically identified in radiology-based screening programs or incidentally when imaging for another injury.⁸⁷

Stress reactions have similar radiological characteristics to pre-bone stress lesions but are symptomatic, highlighting a progression of bone stress continuum, such as more advanced microfractures (Table 29.6). It is possible that the intensity and extent of the signal on MRI may be greater in stress reactions than in pre-bone stress conditions; however, this has not been scientifically validated. At the final stage of the bone stress continuum are *stress fractures*, which are the most advanced bone stress injuries and identified radiologically by the presence of active bone stress (similar to stress reactions) but with a visible fracture line (partial or complete).

Spondylolytic stress fractures are prone to non-union and should therefore be considered as high-risk stress fractures like the navicular and fifth metatarsal stress fractures (Chapter 4). This explains the high rate of non-united spondylolytic fractures (also known as defects) reported in the published literature. A clear distinction between active stress fractures and non-united defects is essential as the management strategies need to be vastly different.

It is therefore important to understand the difference in pathology, clinical presentation and radiological features between active spondylolytic bone pathology and chronic defects.

Clinical presentation

There is no scientific research that outlines the clinical features of spondylolysis. The few papers published highlight a clinical picture that is similar to many other lumbar spine conditions.⁶¹ Kobayashi et al. reported

Table 29.5 Radiological and clinical definition of athletic spondylolytic lesions

Diagnosis	Bone stress (MRI and scintigraphy)	Cortical breach (X-ray, CT, MRI)	Symptoms
Pre-symptomatic bone stress	Yes	No	No
Stress reaction	Yes	No	Yes
Stress fracture	Yes	Yes	Yes
Chronic non-united defect	No	Yes	Yes or no

that key clinical tests were unable to differentiate pain from spondylolytic injury or other back pain.⁶¹ There are also no scientifically validated tests that differentiate spondylolysis between active bone stress injury (stress reaction or fracture) and chronic non-united defects. It is understandable that some clinical features would be similar because the source pain is from the same anatomical location of the posterior vertebral arch.

In active spondylolytic bone stress injuries, pain is typically of gradual onset and often related to changes in training volume and/or repeated cyclical loading of the spine. Pain is localised to the lumbar region, with occasional radiating pain to the buttock and posterior thigh.^{61, 62, 88, 89} Pain often begins as low grade or 'stiffness', but typically progresses (in severity) quickly with continued activity (loading). The pain is typically mechanical in nature, particularly reproduced by spinal side flexion, rotation and hyperextension.^{68, 88-90} Clinical tests that load the ipsilateral posterior vertebral arch and facet joints, such as active and passive lumbar extension and ipsilateral side flexion (or a combination of these movements), reproduce symptoms.^{61, 91, 92} The single-leg extension test or stork test (Fig. 29.3e) had been advocated as a possible clinical test that can specifically differentiate spondylolytic pain from other sources of pain; however, studies have demonstrated that this is not a valid test for spondylolysis.^{93, 94} Localised palpation tenderness and muscle spasm are also common features of spondylolytic injury and may be useful in identifying the specific vertebral level involved.⁹⁴

Imaging

The lack of scientifically validated clinical tests means that radiological imaging has an important role in establishing

the diagnosis when spondylolysis is suspected. It is important to use imaging tests that:

1. differentiate spondylolysis from other sources of back pain
2. confirm the presence of active bone stress pathology
3. establish the stage of the bone stress continuum (Table 29.6 and Fig. 29.5).

What the various imaging modalities provide

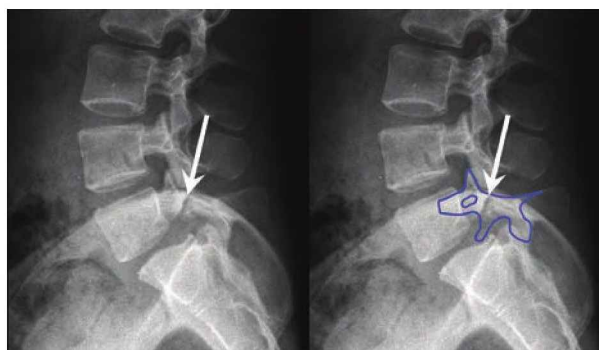
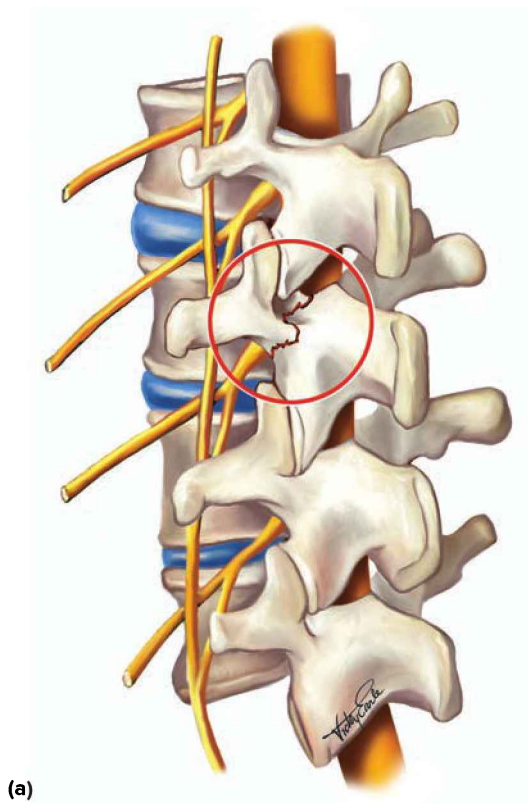
Traditionally, spondylolysis was identified on the oblique radiograph with the classic 'Scotty dog' appearance confirming the presence of a spondylolytic lesion (Fig. 29.5a and b).⁶⁴ However, with advancements in other imaging modalities, plain radiography has a very limited role in detecting spondylolytic lesions because it does not allow differentiation of early active bone stress and does not provide the same level of detail of spondylolytic lesions as other imaging techniques.^{61, 68, 99}

CT (Fig. 29.5c and d), MRI (Fig. 29.5e and f) and scintigraphy (Fig. 29.5g) are therefore the most commonly used imaging modalities when spondylolysis is suspected, with each modality having its advantages and disadvantages in evaluating spondylolytic injury. Sometimes a combination of modalities is required for accurate diagnosis and to differentiate clinically suspected spondylolysis from other sources of back pain.

Bone oedema (high signal on T2 fat saturated sequence) on MRI or increased tracer uptake on scintigraphy (particularly single-photon emission computed tomography [SPECT]) indicates active bone stress injury.^{99, 100} Early comparisons between SPECT and MRI in detecting clinically relevant bone stress had favoured SPECT at detecting early stage bone stress spondylolysis.⁹³ However, with advances

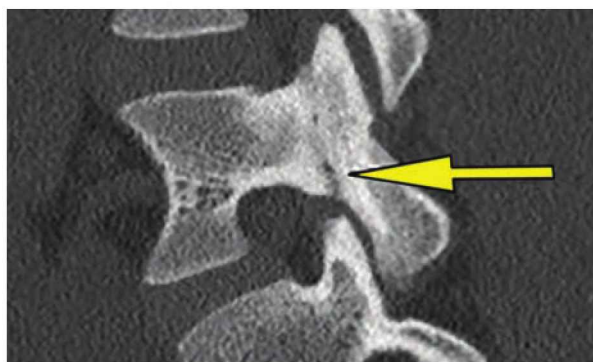
Table 29.6 Imaging appearances along bone stress continuum

	Pre-spondylolytic bone stress	Stress reaction	Stress fracture	Non-united defect
Plain radiograph ^{61, 68}	Unreliable	Unreliable	Unreliable	Scotty dog appearance (oblique view), particularly larger defects (Fig. 29.5a and b)
CT ^{61, 68}	Unreliable—possible bone sclerosis	Unreliable—possible bone sclerosis	Cortical or trabecular fracture (partial or complete) with irregular/ sharp edges (Fig. 29.5c)	Well-defined defect with smooth edges, incomplete ring sign or lysis (Fig. 29.5d)
MRI ^{61, 95, 96}	High-signal bone oedema evident (Fig. 29.5e)	High-signal bone oedema evident (Fig. 29.5e)	Fracture line—complete or incomplete (Fig. 29.5f)	Defect evident
Scintigraphy ^{68, 97, 98}	Increased uptake (Fig. 29.5g)	Increased uptake (Fig. 29.5g)	Increased uptake (Fig. 29.5g)	No increase in uptake

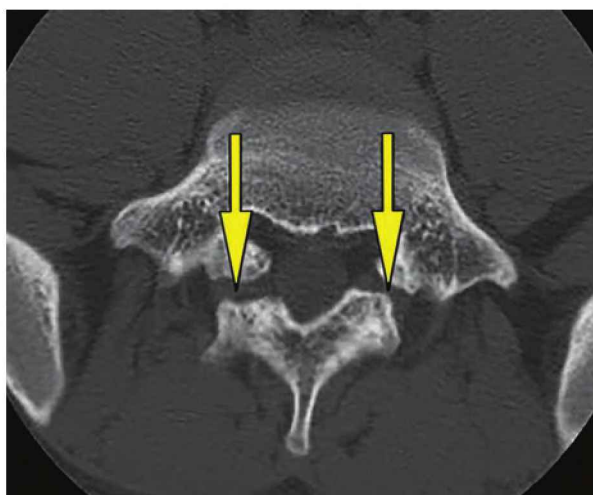


(b)

Figure 29.5 (a) Stress fracture of the pars interarticularis (b) Plain radiograph (arrow) demonstrating the 'Scotty dog' appearance of lumbar pars interarticularis stress fracture (c) Sagittal CT scan of pars interarticularis stress fracture (arrow). Note the possible impingement between the inferior articular facet of the vertebra below and the pars interarticularis of the vertebra above (d) CT scan demonstrating chronic defects (arrows) at the pars interarticularis (e) MRI demonstrating bone oedema (high signal between arrows) at the posterior vertebral arch (b) IMAGING IN SPORTS-SPECIFIC MUSCULOSKELETAL INJURIES 2016, P. 551, ALI GUERMAZI, FRANK W ROEMER, MICHEL D CREMA. © SPRINGER INTERNATIONAL PUBLISHING SWITZERLAND 2016. WITH PERMISSION OF SPRINGER



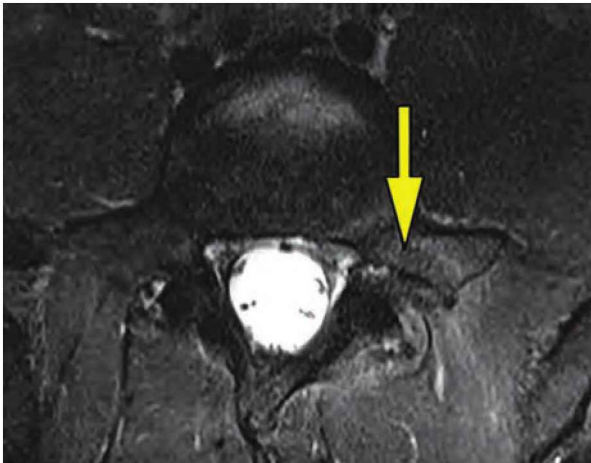
(c)



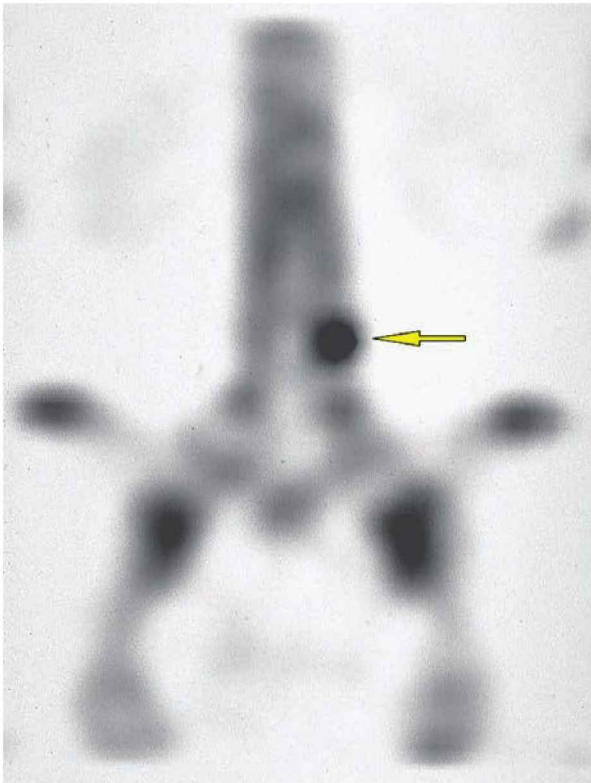
(d)



(e)



(f)



(g)

in MRI imaging technology (higher powered magnets and image sequences) and the associated improvements in radiologists' ability to interpret abnormal signal, MRI is the first choice of imaging modality for detecting bone stress particularly as it does not involve ionising radiation.

Apart from detecting bone stress, MRI can be used to identify the presence of spondylolytic stress fractures,



(h)

Figure 29.5 (cont.) (f) T2-weighted fat-suppressed axial MRI of L5 vertebral body, showing left-sided pars inter-articularis defect (arrow). There is mild bone marrow oedema (bright signal) surrounding the defect, most noteworthy in the left transverse process (g) Scintigraphy (SPECT) demonstrating increased uptake at the posterior vertebral arch (arrow) (h) MRI 3D T1 fat-suppressed image that allows better visualisation of the fracture (arrow)

although thin-slice reverse gantry CT imaging remains the gold standard for imaging these fractures.⁶⁸

CT provides the most detail of a fracture and allows for the grading of spondylolytic lesions as early stage (hairline fractures), progressive stage (wider fractures) or terminal (chronic sclerotic pseudoarthrosis).^{95, 101} The staging of stress fractures is important because early stage lesions have a much better chance of bone healing, compared to progressive lesions, while terminal defects have no ability to heal.¹⁰⁰⁻¹⁰² With advances in MRI, reconstructed fat suppressed T1 MRI images (Fig. 29.5h) can image fractures and non-united defects that would usually require CT imaging.^{95, 96, 98}

The usefulness of MRI extends beyond making the initial diagnosis, with the presence or absence of MRI bone oedema an important determinant of whether bone healing is possible in spondylolytic stress fractures.¹⁰² Sequential MRI imaging following spondylolytic stress fractures has also been used in the management of spondylolysis, with the gradual resolution of bone oedema on monthly scans correlating with bone healing and the resolution of pain.¹⁰⁰

As discussed earlier, non-union is a common outcome following spondylolytic stress fractures in athletic populations.^{85, 103} Unlike some other stress fractures such as the navicular and fifth metatarsal, athletes with non-united spondylolytic stress fractures can remain asymptomatic and return to playing, even at the elite level.^{85, 103} MRI, plain radiograph and CT imaging can detect chronic defects as sclerotic scar-filled clefts. These fibrous defects are prone to becoming symptomatic with mechanical loading, but the pain is similar to that from other nociceptive structures such as the intervertebral discs or facet joints. To determine if the defect is the source of pain, the gold standard is radiologically guided local anaesthetic injection into the defect and re-assessing key clinical tests.⁹⁰

Pathogenesis

As with other bone stress injuries, the pathogenesis for athletic spondylolysis involves localised repetitive bone loading that consistently exceeds the normal bone strain threshold.^{60, 104, 105} The posterior vertebral arch is loaded during trunk extension, rotation and side flexion movements.¹⁰⁶ It is possible that these movements either lead to shear forces or bony impingement that result in microfractures (early bone stress) and eventually lead to symptomatic stress (macro) fractures.¹⁰⁶⁻¹⁰⁸

High ground reaction forces contribute to athletes exceeding the bone strain threshold. For example, cricket fast bowlers can have weekly bowling loads of 150 to 300 deliveries or 5000 to 7000 deliveries per year, with ground reaction forces of 7-10 times body weight per delivery.¹⁰⁹

The magnitude and frequency of load required to fracture the posterior vertebral arch is unknown, but individual variations (intrinsic factors) may predispose some athletes to lower fracture thresholds than others. Age influences the fracture threshold for spondylolytic bone stress, with younger and skeletally immature athletes at greatest risk.⁷² Bone geometry is also important as a lower proportion of cortical to trabecular bone at the pars region increases risk of fracture.¹¹⁰ Other intrinsic factors such as shorter interfacet distances,^{105, 111} increased coronal facet joint orientation,¹¹² excessive lumbar lordosis^{105, 111, 113} and increased pedicle length¹¹⁴ are associated with spondylolytic injury.

As discussed previously, spondylolytic stress fractures are prone to non-union, which results in a scar-filled defect and eventually a pseudoarthrosis.^{60, 95, 100-102, 115} The reason for the high rates of non-union are unknown, but likely to be related to either an inadequate period of unloading to allow healing, or because of a delay in confirming the diagnosis (continue playing sport) that may disrupt the bone's attempt at healing. The non-united defect eventually fills with scar tissue to form a pseudoarthrosis.

The histological appearance of the scar tissue is that of a fibrocartilaginous mass that has similar properties to ligamentous tissue combined with a small amount of osseous particles, and has been described as enthesis-like, resembling a ligament-bone junction.¹¹⁵ The enthesis-like nature of the spondylolytic mass allows it to stabilise the vertebral arch and absorb tensile forces.¹¹⁵ This enthesis-like material suggests that pain from spondylolytic defects can be analogous to a joint or tendon pathology and should be managed in the same manner once symptomatic.

When non-union occurs and enough time is allowed for the fibrocartilaginous mass to mature and stabilise, some athletes can return to an elite level of athletic participation.⁸⁵ It is not clear why some athletes with chronic defects cope and others do not, and why some defects remain asymptomatic and others painful. It is likely that repetitive mechanical loading or single high-force trauma (e.g. landing in a lumbar hyperextension) causes disruption of the fibrocartilaginous tissue, leading to pain.

PRACTICE PEARL

The pseudoarthrosis and nature of the enthesis-like mass suggest that symptomatic defects should be managed as chronic (or acute on chronic) joint or tendon injuries.

In rare cases, herniation of the fibrocartilaginous mass can cause nerve root impingement and radicular symptoms.¹¹⁵

Management of lumbar spondylolysis

The difference in the pathogenesis of spondylolysis between symptomatic bone stress injuries (stress reactions and fractures) and non-united defects highlights the importance of accurate diagnosis and that management strategies need to be appropriate for the pathology.

PRACTICE PEARL

Management of spondylolysis needs to be considered as two separate entities: active bone stress injury and symptomatic non-united defects.

Active spondylolytic bone stress injury

The key management goals for active spondylolytic bone stress injury (stress reactions and stress fractures) are the resolution of symptoms, bone healing, return to pre-injury function and avoidance of recurrence. All four goals are interrelated and bone healing is key to the process.^{86, 89, 116, 117}

Conservative management can produce excellent clinical and radiological outcomes, particularly in young athletic populations.^{86, 88, 118-123} However, return-to-sport times range from 3-6 months and complex cases take up to 12 months.^{86, 89, 101, 102, 121, 123, 124} This reflects the physiology of bone healing and correlates with imaging resolution of bone oedema on MRI and bone union on CT.¹⁰⁰ It reflects the need to invest time in a well-planned rehabilitation program that addresses musculoskeletal and biomechanical deficiencies.

The type of exercise and loading activities should be guided by clinical symptoms, bone healing times, imaging and clinical assessment. To simplify this, the rehabilitation program can be divided into three distinct periods that have specific focus.

Weeks 0-8: fracture healing phase

The aim is to protect the stress fracture and create an environment to enhance bone healing. For stress reactions, the aim is to prevent the progression to a stress fracture, and for stress fractures it is to attain bone healing and avoid non-union.

The most important aspect of the early management is to cease sporting activities that load and stress the posterior vertebral arch, such as running, jumping and landing. It also includes ceasing other training activities such as resistance-based strength training that involves axial loading¹²⁵ and avoiding repeated or end-range trunk positions, particularly extension, rotation and side flexion that may load the posterior elements of the vertebrae.¹⁰¹

It is not clear whether any specific exercises can improve fracture healing, reduce recurrence rates or enhance return-to-sport times, but when planning rehabilitation, prescribe exercises that can minimise muscle atrophy, address lumbopelvic deficiencies and improve the lower limb kinetic chain deficiencies that are often present in the young athletes with spondylolysis.⁷¹ In particular, prescribe exercises that focus on strength, control and flexibility of the lumbopelvic region, such as abdominal strengthening and 'core stabilising'.¹²³ Lumbar extensor muscle endurance has been linked with the development of lumbar bone stress injury in cricket fast bowlers and should be considered in those identified to have insufficient strength and endurance.¹²⁶

PRACTICE PEARL

A good starting point with the rehabilitation program is lower limb body weight exercises such as squatting, bridging and lunging to address strength and control deficits, while minimising excessive lordosis and developing a base for strength training later in the rehabilitation program.

Exercises to improve lumbopelvic control are also appropriate to minimise the effects of deconditioning of the trunk postural muscles, particularly in athletes with ongoing pain. It is important to closely monitor clinical signs during this part of the rehabilitation program, as any reproduction of symptoms during or after the exercise could be a sign of disruption of the fracture site. General exercise such as walking and cycling is safe and can be used to maintain cardiovascular fitness.

The use of thoracolumbar bracing can be considered during this stage to restrict spinal movements during daily activities and protect the fracture. There are some published reports that demonstrate good clinical and radiological (bone healing) outcomes with the use of strict protocols for bracing following lumbar bone stress injuries.^{71, 100, 123} The brace needs to be worn almost all day and for months to enhance the potential for healing. A practical approach to brace treatment might be to use it in athletes who have pain at rest or with daily activities during the early period post-injury, in those with excessive lumbar lordosis, if there is radiological evidence of delayed bone healing, or with bilateral stress fractures.^{71, 123}

Weeks 9-16: protected reloading phase

During this stage, there should be total resolution of key clinical signs, while daily activities and rehabilitation exercises should also remain pain-free. The introduction of low-impact activities that specifically load the lumbar spine can commence and theoretically stimulate bone healing through the process of mechano-transduction.¹²⁷

Typically, the bone oedema will be less apparent on MRI during this stage, indicating resolution of bone stress, and in most cases should have totally resolved by 12 weeks post injury.¹⁰⁰ It is unknown how MRI bone oedema correlates with physiological bone healing, but Sakai et al. demonstrated that the resolution of bone oedema in active spondylolytic injury occurred between 3-4 months post injury and correlated well with bone healing observed on CT imaging and resolution of symptoms.¹⁰⁰

The progression from body weight to loaded strength exercises such as dumbbell leg squats and lunges is

an appropriate transition, while lumbopelvic control exercises can also be progressed to more functional trunk positions. Body weight exercises involving trunk rotation and extension can commence to gradually load the spine and stimulate healing.

Jogging can commence during this period with a gradual transition to higher intensity running by (approximately) week 12 onwards if symptoms and imaging suggest that healing is progressing without complication. In the latter part of this stage, functional submaximal dynamic exercises (e.g. plyometric) or sports-specific drills can commence at lower intensity and volume. For example, hopping or jumping activities for athletes involved in sports such as basketball and volleyball, kicking drills for sports such as football (soccer) and Australian football, or drills that mimic specific activities such as cricket fast bowling can commence.

The commencement and progression of exercises and activities should be guided by consistent absence of localised LBP and no change in clinical symptoms on key tests. If there is any deterioration in clinical presentation, the progression of exercises should be reviewed until symptoms resolve. Additionally, if imaging demonstrates that bone healing is delayed, these milestones should be held back until satisfactory progress is evident.

Week 17 onwards: transition to return to full function and sport

This is an important and often neglected part of the rehabilitation process that should involve careful monitoring of symptoms, reassessment of key physical competencies, addressing biomechanical changes and gradually introducing unrestricted load until there is a full return to pre-injury function. All daily activities, training exercises and key clinical tests should remain pain-free during this stage.

Radiologically, evidence of advanced bone healing would be expected during this stage. In particular, resolution of MRI bone oedema has been demonstrated by 16 weeks post injury and coincides with bone healing on CT imaging.¹⁰⁰ Repeating an MRI (fracture views shown in Fig. 29.6) and/or limited (only the appropriate vertebral segment) CT scan at approximately 16 weeks post injury is useful to determine the bone healing response.

PRACTICE PEARL

Although radiological healing is an encouraging sign in the recovery process, it is unknown how this correlates with physiological bone healing.

The key rehabilitation principle for this stage is the gradual increase in rehabilitation and training loads to allow a safe transition to return to play. As part of the process, unrestricted strength training can be progressed to ballistic exercise. Running and other conditioning methods can become sports specific, but it is important to monitor both workload intensity and volume during this stage, so the transition from the training ground to playing arena is graduated appropriately avoiding 'spikes' in workloads, that have been shown to be linked with athletic injuries.^{128, 129} Sport-specific exercise and training is gradually progressed so that intensity and workload are at a level required to return to competition. Once again, an exacerbation of LBP in the region of the previous injury and reproduction of pain with key clinical tests should be seen as a sign to slow down or to cease the activity.

Symptomatic non-united fractures or chronic defects
When spondylolytic stress fractures fail to heal, the remaining defect fills with scar tissue that offers enough

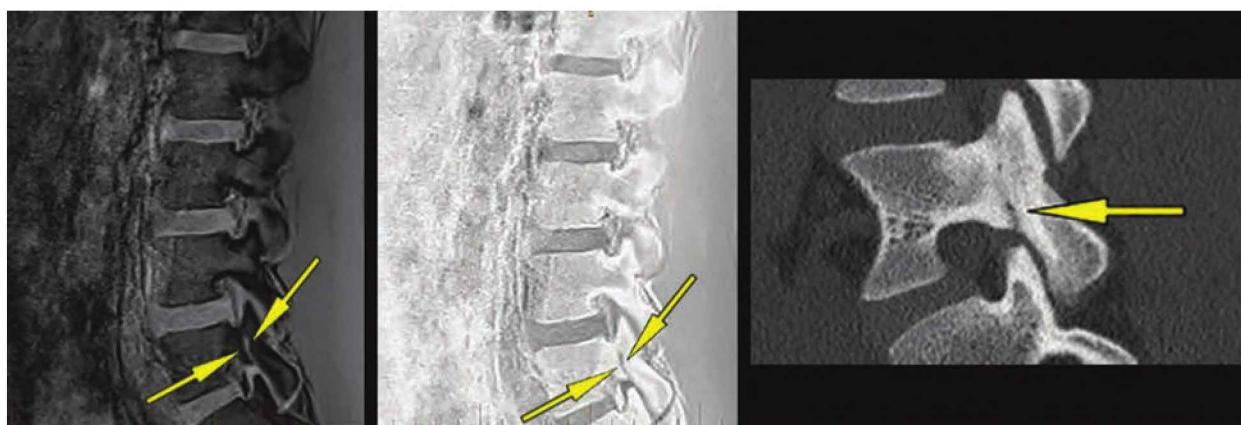


Figure 29.6 Different imaging modalities. The image on the left is an MRI T1 fat-suppressed three-dimensional image. The second is the same MRI image inverted (black becomes white) to make it look similar to CT. The third is the CT image of the same patient on the same day. Note the similarity between the CT and MRI images

stability to the segment to allow some athletes to return to unrestricted pre-injury function, even at the elite level. It is unknown how many athletes with non-united spondylolytic stress fractures fail to return to pre-injury level of activity, but there are numerous reports of athletes competing at the elite level with (often asymptomatic) chronic lumbar defects.⁸⁰ If an athlete does return to sporting activities with a non-united defect, episodic or ongoing pain is common if the fibrous scar tissue is disrupted and irritated with mechanical loading.

An exacerbation of pain from a non-united spondylolytic defect needs to be managed as an acute or chronic injury. The acute component is likely to be inflammatory in nature due to mechanical disruption of the fibrous scar and can respond to appropriate anti-inflammatory management strategies. The chronic nature of the defect does mean that there is virtually no possibility of bone healing, so this should not be considered as a viable management goal.^{101, 102, 121}

Symptom management

With little prospect of bone healing, the main aim of treatment is to reduce pain and restore function. While the exact source of pain is unknown, disruption and inflammation of the fibrous scar is a likely mechanism so the use of these anti-inflammatory methods should be considered as well as analgesic medication and possibly oral corticosteroids.

Radiologically guided corticosteroid and local anaesthetic injection within the defect (Figure 29.7) has considerable merit. The short-acting local anaesthetic is useful in determining if the defect is the source of pain, while the corticosteroid can provide the localised anti-inflammatory response thereby reducing pain and allowing a faster transition through the rehabilitation program and return to function.⁹⁰

Unloading phase

The nature of the fibrous tissue within the bony defect, particularly that it is histologically similar to ligament

and tendon, suggests that the management of non-union should be similar to the management of those tissues. Importantly, a brief period of unloading to allow irritability of the fibrocartilaginous scar to resolve may be required, with rest from aggravating (sporting) activities required if pain does not settle. The length of the unloading period needs to be guided by the nature of the pain and typically involves initially reducing exposure to provocative activities that involve repetitive trunk extension, rotation and side flexion or any other aggravating activity.

The length of the unloading phase and the transition to reloading should depend on the reduction of symptoms and improvement in painful clinical tests.

PRACTICE PEARL

Unlike acute bone stress injuries, total resolution of pain of non-union is not necessary, but careful monitoring of response to reloading is important.

It may be possible that athletes can return to sport with some low-grade pain as long as the pain characteristics remain manageable with increase in load and during the transition to sporting function. In this sense, the unloading phase for chronic symptomatic pars defects is similar to tendinopathy, where the unloading phase serves to reduce but not always eliminate pain.¹³⁰

Summary

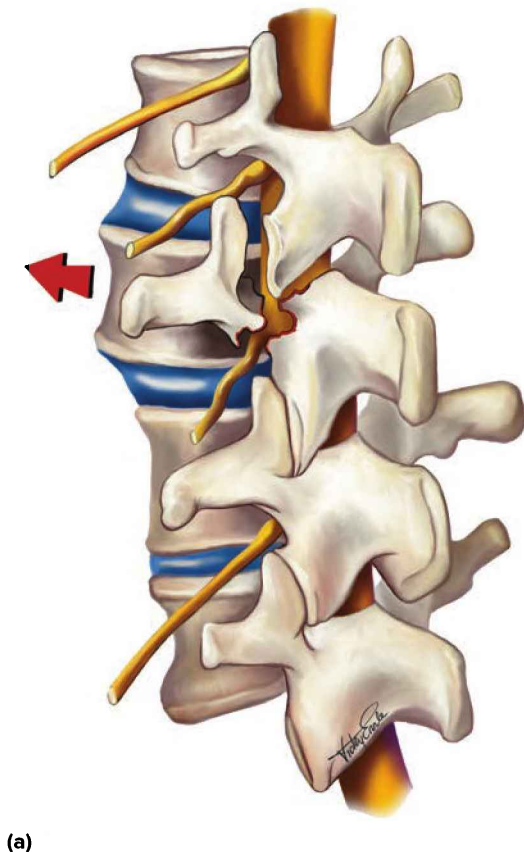
Lumbar spondylolysis is a complex condition with little scientific evidence underpinning the clinical tests to differentiate it from other causes of lower back pain. Even more complex is understanding the stage in the bone stress continuum as this is important in the management of spondylolysis. Advances in imaging have improved the understanding of this condition, particularly the stage in the bone stress continuum and differentiating active bone stress injury from symptomatic non-united defects. The management of spondylolysis needs to consider the clinical features and the imaging. The management of active spondylolysis needs to focus on bone healing that takes months, while symptomatic non-united fractures can be treated symptomatically, reducing the recovery period.

Spondylolisthesis

Spondylolisthesis (Fig. 29.8a) refers to the anterior displacement of part or all of one vertebra forwards on another. It is often associated with bilateral pars defects that usually develop in early childhood and have a definite family predisposition. Pars defects that develop due to athletic activity (stress fractures) rarely result in spondylolisthesis.



Figure 29.7 Guided injection into the chronic pars defect



Spondylolisthesis is most commonly seen in children between the ages of 9 and 14. In the vast majority of cases, it is the L5 vertebra that slips forwards on the S1. The spondylolisthesis is graded according to the degree of slip of the vertebra. A grade I slip denotes that a vertebra has slipped up to 25% over the body of the vertebra underlying it; in a grade II slip the displacement is greater than 25%; in a grade III slip, greater than 50%; and in a grade IV slip, greater than 75%. It is rare for a slip to progress once the athlete has reached maturation. Lateral X-rays best demonstrate the extent and possible progression of vertebral slippage (Fig. 29.8b). A recent systematic review suggests that spondylolisthesis is not predictive of LBP in the general population.¹³¹ However, in an athletic population it may become symptomatic.

Clinical features

Spondylolisthesis is often asymptomatic and the patients may be unaware of the defect. However, LBP, with or without leg pain associated with the spondylolisthesis, may develop in response to various biopsychosocial factors and is frequently aggravated by extension activities and postures. On examination, there may be a palpable step off corresponding to the slip.



Figure 29.8 Grade I spondylolisthesis and pars defect (a) Anterior slip (arrow) (b) Plain radiograph (single arrow shows the defect; dotted lines show the method of measuring slippage)

Treatment

In considering the treatment of this condition, it is important to remember that the patient's LBP is not necessarily being caused by the spondylolisthesis. First, it is important to reassure the athlete that spondylolisthesis is common in the pain-free population and the natural history for these disorders is usually very good. Addressing the various biopsychosocial features of the disorder described above is central to management.

Treatment of athletes with grade I or grade II symptomatic spondylolisthesis is usually non-surgical, and rehabilitation exercise approaches are shown to be effective.¹³² Athletes with grade I or grade II spondylolisthesis may return to sport after their rehabilitation program, when they have minimal pain on extension and have good spinal control during functional and sports-specific activity. If the symptoms recur and are distressing, activity should be modified.

Athletes with grade III or grade IV spondylolisthesis should avoid high-speed or contact sports. Treatment is initially conservative, yet if a patient continues to experience symptoms after a 6-month non-surgical treatment period, he/she may be considered for surgical fixation. Other indications for surgery include growing children with a

slip >50%, progressively worsening neurological signs and radiological evidence of further displacement.¹³³

Acute radiculopathy +/- nerve root compression

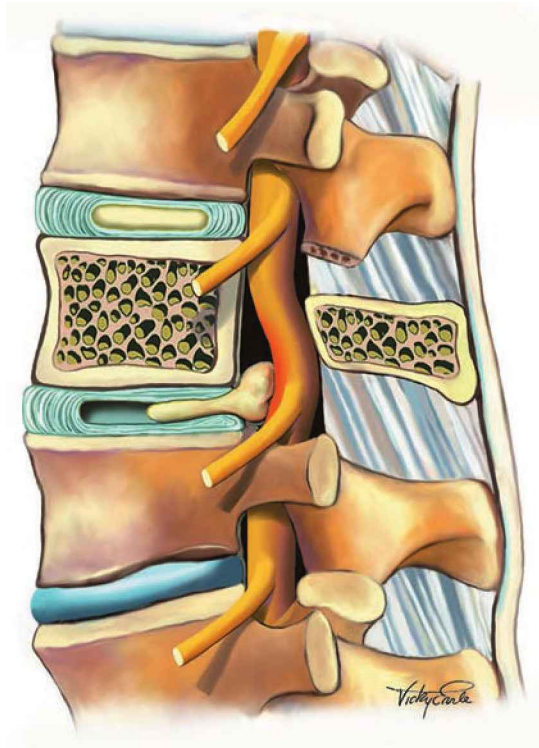
Acute radiculopathy is usually the result of an acute disc prolapse when the contents of the nucleus pulposus of the intervertebral disc are extruded through the annulus fibrosis into the spinal canal, irritating the nerve root (Fig. 29.9). The irritation of the nerve root may be due to direct mechanical compression by the nuclear material and/or as a result of the chemical irritation caused by the prolapse, and is usually very painful and distressing.

Prolapse usually occurs in discs that have been previously weakened by one of the processes mentioned already. Disc prolapse usually occurs between the ages of 20 and 50 years and is more common in males than

females. The L5-S1 disc is the most commonly prolapsed disc and L4-5 the next most common.

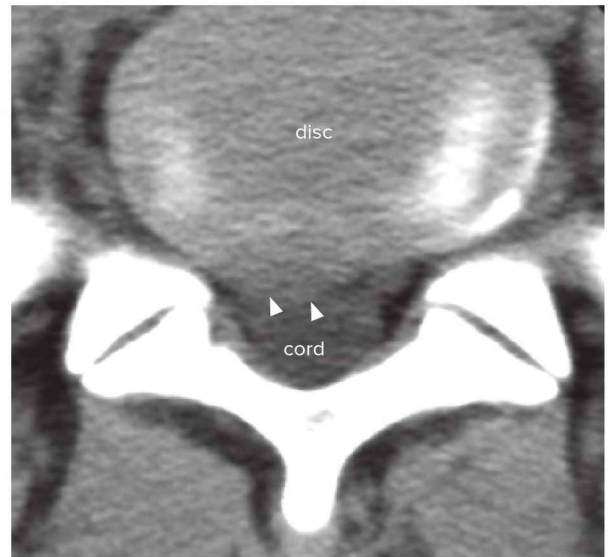
Clinical features

Typically, a patient with a disc prolapse presents with acute LBP and radicular leg pain following a relatively trivial movement, usually involving forward bending. This is frequently preceded by a period of rapid increase in repeated flexion loading (such as a sudden increase in rowing training volumes). On occasion, the presentation may be painless, with weakness or sensory symptoms only (in the case of nerve compression).



(a)

Figure 29.9 Disc extrusion compressing nerve root (a) Spinal cord compression from severe disc prolapse (b) Axial CT scan. Arrowheads point to the border between the bulging disc (anterior) and the compressed cord (posterior) (c) Sagittal MRI showing prominent posterior L3-4 disc bulging (arrow) and cauda equina compression. The L4-5 disc (arrowhead) is also abnormal (compressed) but not protruding



(b)



(c)

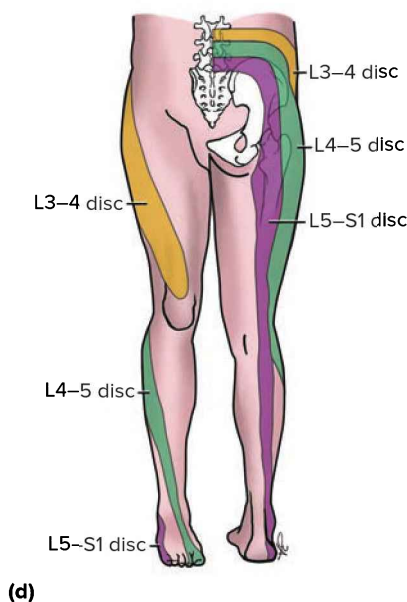


Figure 29.9 (cont.) (d) Common areas of pain radiation with disc prolapse at the L3-4, L4-5 and L5-S1 levels

The symptoms depend on the direction of the prolapse. Posterior herniations are more likely to cause LBP with later development of leg pain, whereas posterolateral herniations may cause radicular symptoms without LBP. Sharp, lancinating pain in a narrow band down the leg is common with radicular pain and is associated with nerve root irritation and positive SLR, accompanied by pins and needles, numbness and weakness. Pain is often aggravated by sitting, bending, lifting, coughing or sneezing. Pain is usually eased by lying down, particularly on the asymptomatic side, and is often less after a night's rest.

On examination, the patient often demonstrates a list to one side that is usually, although not always, away from the side of pain (Fig. 29.10). This is a protective scoliosis and reflects an adaptive motor response. Examination may be difficult if there is severe pain and irritability. Straight leg raise is usually limited (less than 30° in severe cases) and all active movements of the lumbar spine, particularly flexion, are usually restricted by muscle guarding. Palpation usually reveals muscle spasm with marked tenderness but occasionally it may be unremarkable. A neurological examination should always be performed when pain extends past the buttock fold or there are subjective sensory and/or motor changes.

Treatment

It is important to educate the patient that the natural history for disc prolapse is excellent for the majority, with resolution of symptoms and reabsorption of the prolapse occurring



Figure 29.10 List to the left, commonly associated with pain on the right

over a 3-6 month period of time. The 2-year outcomes for surgical compared to non-surgical interventions are the same.¹³⁴ However, any deterioration in bladder function or neurological deficits must be monitored and reported immediately for investigation and surgical review.

In the acute phase (1-5 days), if pain is severe, adopting positions of maximum comfort with administration of analgesics and NSAIDs is indicated. The patient should minimise sitting and keep active with gentle walking. In the absence of nerve compression (neurological signs), gentle spine mobility exercises such as extension and/or rotation exercises (Fig 29.11) may provide pain relief and if so should be commenced as soon as possible. However, if exercises cause an increase in peripheral (i.e. lower extremity) symptoms and/or neurological signs, they should be ceased immediately.

Mobilisation techniques should be performed with care. Rotational positioning (Fig. 29.11) may be effective in providing pain relief. Manipulation is contraindicated in conditions with severe pain, muscle guarding and acute neurological signs and symptoms.



Figure 29.11 Rotation exercise for acute LBP

A transforaminal epidural injection of corticosteroid may help if there is no significant improvement in symptoms and signs with rest, and pain levels are distressing. Selective nerve root blocks using only local anaesthetic can be helpful to confirm a particular nerve root as a source for radicular leg pain. A transforaminal epidural steroid injection can be performed to provide pain relief from the implicated irritated nerve root. These injections are more specific and concentrate the medication around the nerve root as compared to interlaminar or caudal epidural steroid injections that cover a larger area yet place a more dilute injectate at each level.¹³⁵

Epidural injections with a long-acting local anaesthetic +/- a corticosteroid are a treatment for radicular leg pain and not a treatment for LBP.¹³⁶ However, there is little research evidence for the efficacy of these procedures in chronic radicular pain.

As the acute episode settles, restoration of normal pain-free movement to the area can be assisted with localised mobilisation and gentle stretching. Following restoration of range of movement, optimising patterns of spinal control, conditioning and adjustment of sporting technique is important in order to optimise stress on spinal structures and prevent recurrence. This is discussed later in this chapter.

Spinal surgery should only be considered when nerve root compression results in persistent: bladder or bowel symptoms, progressive sensory or motor abnormalities, or where radicular pain is intractable despite adequate conservative management. Various surgical options exist including microsurgical decompression, open laminectomy or percutaneous discectomy using a needle aspiration technique.

While complications are rare, it is important for the clinician to advise the patient of the potential risks for each of the interventions just discussed, which include infection, bleeding, worsening pain, allergic reaction, vascular and neural injury, as well as intravascular injection.¹³⁵

Vertebral endplate oedema (Modic type 1 changes)

A patient with vertebral endplate oedema typically presents with persistent, severe and disabling localised LBP. Pain is usually constant (at rest and at night) and exacerbated with loading and movement. Diagnosis is based on MRI with finding of oedema of the vertebral endplate proximal to a degenerate disc (Fig. 29.12).

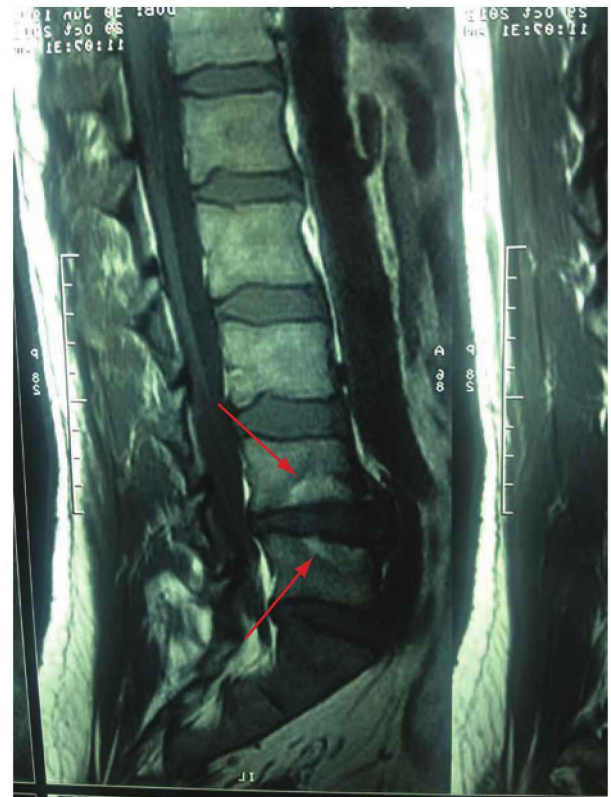


Figure 29.12 MRI demonstrating disc degeneration and moderate intensity signal Modic type 1 changes (arrows)

Treatment

Management is consistent with other LBP disorders (outlined in the next section) and involves addressing the relevant extrinsic and intrinsic risk physical factors, as well as lifestyle and psychosocial factors in conjunction with medical management (anti-inflammatory medication). A period of relative unloading from sport, combined with regular unloaded cardiovascular exercise (water-based/exercise bike), is usually required to allow the inflammatory process to settle, followed by a gradual return to sport. The natural history is good.

There is some preliminary evidence to suggest that these MRI findings, when presenting secondary to a disc prolapse, may represent an underlying low-grade infection. One randomised controlled trial (RCT) reported benefit with treating this using broad-spectrum antibiotics, although replication trials are required to confirm this.¹³⁷

MANAGEMENT OF NON-SPECIFIC LBP DISORDERS

Once the patient has been triaged to exclude serious and specific pathology (Fig. 29.2), treating the symptom of pain when it is acute, severe and distressing is very important. However, there is growing evidence to support that targeting the modifiable beliefs and behaviours that drive pain and disability may be more effective than simply treating the symptom of pain in order to achieve long-term changes and especially where LBP is persistent and/or recurrent.^{16, 138}

An integrated person-centred, goal-orientated management approach for LBP is adapted to the time course and complexity of the patient's profile. The focus of this process is directed by the findings taken from the examination with regard to the primary contributing factors, across the different domains, linked to the patient's disorder.

PRACTICE PEARL

Developing a strong clinical alliance underpins the therapeutic process. Effective communication encompassing therapist empathy, active listening, responsiveness, validation of the patient, confidence, warmth and friendliness are central to clinical outcomes.

Any management approach in the athlete must be applied in discussion with the coach and in consideration of managing training loads.

Clinical approach

Effective patient education

As LBP can often be distressing and associated with increased fear, catastrophising, vigilance, anxiety and worry, providing the athlete with a clear understanding

of the underlying mechanisms linked to the disorder and directions for treatment and management is central to any intervention. This may involve:

- providing effective patient-centred education regarding the multidimensional mechanisms that drive their pain and disability, as well as clear expectations for prognosis and management
- education regarding the influence of beliefs, fear, vigilance, levels of mood, anxiety, stress, worry, sleep deficits and levels of fatigue on pain physiology
- addressing negative beliefs and fear regarding pain with positive information regarding the spine's resilience and providing epidemiological advice regarding MRI findings if relevant²¹
- education regarding the importance of optimal motor control, body relaxation and conditioning, and the interplay this has with sports-specific biomechanical demands and training loads to influence tissue strain and sensitivity
- promoting active coping strategies for pain and installing confidence
- facilitating goal-orientated behavioural change regarding stress management, sleep, physical activity, pacing and diet
- feedback, which is critical and involves:
 - mindfulness of the body's responses to pain and
 - visual feedback with the use of mirrors, video and written instruction.

PRACTICE PEARL

Effective, evidence-based patient education and targeted reassurance is a treatment in its own right. (See also Chapter 5 which underscores that explaining pain is a treatment in its own right.)

Pain relief

There is strong evidence that symptomatic treatments for LBP (such as manual therapy, soft tissue therapy, needling, spinal injections and pharmacology), while providing a window of short-term pain relief, have little impact on the long-term trajectories of LBP. Therefore, these treatments need to be closely integrated with active management strategies that address the relevant biopsychosocial factors identified in the examination. They should be seen as creating a window of opportunity to enhance functional capacity, but should not be relied on as a treatment in itself. Furthermore, creating a reliance on passive therapies may be unhelpful in the long term as it likely reduces the patient's levels of self-efficacy and internal locus of control.

Functional restoration

Restoration of full range of relaxed spinal movement is the first priority. Once this has been achieved, then

pain-provocative postural and movement patterns and protective behaviours are identified (specific to the individual), broken down into component parts and retrained in a mindful and relaxed manner in order to provide strategies that enable the athlete to actively control their pain during functional tasks (see later section).

These 'new' postural and movement patterns are gradually targeted towards the activities and movements that provoke pain and/or are avoided by the patient in order to reduce the threat value of the task and normalise it. These 'new' patterns are then integrated into the patient's daily life and sporting activities in a graduated manner. Targeted strengthening and conditioning is incorporated as required by the functional goals of the patient and the specific demands of the sport.

Addressing lifestyle aspects

Addressing lifestyle factors such as sleep hygiene, body-mind relaxation (using meditation techniques), activity pacing, healthy diet, reducing sedentary behaviours, alcohol consumption and drug taking where relevant is also important.¹⁷

Multidisciplinary care

In situations where pain is severe and a barrier to rehabilitation, integrating care with medical management may be indicated. Where psychological comorbidities dominate, combined psychological management may be required.

Acute severe low back pain

Acute severe LBP associated with a clear history of a significant sudden or repeated high-level biomechanical loading of the lumbar spine, where there is no sign of disc prolapse and radiculopathy or risk of fracture, should be considered an acute tissue strain. The pain response may involve local sensitisation and an inflammatory sequel related to affected spinal structures, which may involve the vertebral endplate, intervertebral disc, zygapophyseal joints and/or associated ligamentous structures.

PRACTICE PEARL

As clinical examination cannot accurately determine the specific source of pain, a 'structure specific' diagnosis cannot be made with any confidence. Reassuring the athlete that these 'tissue strains', while very painful, have an excellent natural history if managed well, is important.

However, it must also be noted that severity of symptoms and levels of distress for acute LBP are often a poor indicator of levels of tissue damage, and acute pain may also reflect the level of underlying tissue sensitisation

and contextual distress of the individual. This may be characterised by an athlete who presents with acute pain and distress linked to a minor mechanical loading event or an insidious onset of pain. Careful questioning commonly identifies high levels of contextual stress, anxiety or mood fluctuations coupled with lifestyle factors (such as sleep deficits and fatigue) as precipitating factors in these presentations.

These disorders should not be treated as an 'injury', with care being directed to reassure the person that they aren't 'damaged', but that their spinal structures are sensitised by contextual factors in their life. These disorders usually settle quickly with confident management that targets reassurance, body relaxation and functional restoration while addressing contextual stress and lifestyle-related factors. Careful correlation between the history of pain (i.e. injury mechanism/s), lifestyle factors, contextual stresses, psychosocial screening and disability levels can help differentiate these groups.

Clinical features

The patient presents with high ratings of pain and disability of sudden onset. The pain is usually in the lower lumbar area, but it may radiate to the buttocks, hamstrings or lower leg, and is somatic in nature, with the patient complaining of a deep-seated ache. The patient with acute, sudden onset of LBP often adopts a fixed, guarded position and movements are severely restricted. Palpation of the lumbar spine reveals areas of marked tenderness with associated muscle spasm.

Management

Successful management of acute severe low back pain requires education, pain control and progressive rehabilitation to permit the patient to regain function.

Education

Once triage has been completed to rule out serious and specific pathology, educate the patient that pain is associated with acute sensitisation of spinal structures. Reinforce that levels of pain do not equate well to degree of damage and that a severe pain experience can be related to sensitisation of spinal structures, amplified by sleep deficits, contextual stress, lower mood, heightened anxiety, fear and muscle guarding, and so on. Keeping the athlete calm, relaxed and positive is central to effective management. Reassure the patient that these disorders are very painful, but self-limiting and with an excellent natural history.²¹

Pain control and functional restoration

Initially, encourage the patient to adopt the position of most comfort and relax using diaphragm breathing. This position is often supine crook lying depending on the



Figure 29.13 Taping

patient's pattern of sensitisation. Following this, these management strategies may be helpful:

- Movements that aggravate pain should initially be limited, whereas movements that reduce or minimise pain should be encouraged.
- Relative rest in the position of most comfort may be continued for up to 48 hours if pain is severe

and disabling and related to biomechanical loading. Bed rest longer than 48 hours has been shown to be detrimental.¹³⁹

- Taping of the low back (Fig. 29.13) for the first 48 hours can reduce acute back pain, provide reassurance, allow trunk muscle relaxation and allow quicker functional restoration.
- Analgesics may control the pain and reflex muscle spasm. NSAIDs may help reduce inflammation if a clear pattern of tissue injury is present.
- Exercise in a direction away from the movement that initially provoked the patient's symptoms should be commenced as early as possible.

For example, for those patients in whom lumbar flexion aggravates their symptoms (linked to flexion strain), extension exercises can be performed. The degree of extension should be determined by the level of pain sensitivity and pattern of pain provocation. Initially, lying prone may be sufficient. Later, extension of the lumbar spine by pushing up onto the elbows may be possible. Eventually, further extension with straight arms can be achieved.

In patients for whom extension movements aggravate their pain (usually linked to an extension pattern of strain), flexion exercises can be performed (Fig. 29.14). For these patients, prolonged posture involving extension, such as standing with excessive lumbar lordosis, should be minimised in the acute phase (48 hours).

If a side bending or rotational strain has occurred, then rotation and side bending away from the direction of tissue strain and pain provocation can be performed.

PRACTICE PEARL

Exercises should be discontinued if peripheralisation or exacerbation of symptoms occurs (i.e. increasing pain in the lower extremity).



Figure 29.14 Flexion exercise

Manual therapy has a limited role in treating severe LBP where there has been a clear history of tissue trauma. Gentle mobilisation techniques, such as non-provocative physiological movements of the spine, may assist the patient to relax and move; however, the patient's response must be monitored. If there is deterioration of symptoms, mobilisation should be ceased. The mobilisation should be performed in the position of comfort adopted by the patient and in the direction of pain 'easing'.

Similarly, gentle soft tissue therapy may be helpful in relieving pain and muscle spasm. Traction has not been found to be helpful in patients with acute LBP. Manipulation should not be attempted in the presence of marked muscle spasm.

In the absence of a clear history of tissue strain, manual therapy can be very helpful to relax the person and facilitate functional restoration in a graded manner and with respect to pain.

Subacute low back pain

Once the acute phase (up to 48 hours) of severe LBP has passed with reduction in pain and muscle spasm,

graduated exercise therapy can be commenced. The aim of this stage of therapy is to first restore normal relaxed spinal movement. This can be facilitated with the use of pain-relieving manual and soft tissue treatments (outlined below) in conjunction with exercise therapy.

Once normal spinal movement has been restored, this is gradually integrated into functional postures and movements of daily living with attention to confident relaxed body control and other multidimensional factors associated with the onset of the pain disorder based on their complexity profile (see below). Once the person's symptoms settle and spinal movement is normalised (usually within 1-3 weeks), then a graduated return-to-sport program is developed.

The over-arching aims are to provide a clear understanding to the patient of their disorder, enhance pain controllability and return the athlete to their desired level of activity with confidence. An exacerbation prevention and management plan is also important where recurrent back pain is a feature of the disorder. To monitor the effectiveness of the treatment, regular screening questionnaires are helpful in order to monitor levels of pain, disability and distress.

Passive treatments for low back pain

Passive treatments for LBP (where the patient is the recipient of the treatment rather than being actively involved) may be helpful where pain acts as a barrier to functional activation. These treatments can be used as a window of opportunity in the acute or subacute phase to provide pain relief, to activate the athlete. It is important for the clinician to recognise that passive therapies only provide short-term pain relief and they do not change the natural history of the disorder on their own.

Where active pain control strategies can be integrated with these treatments, such as mindfulness and relaxation techniques as well as exercise, functional activation and lifestyle modifications, they should be actively encouraged. Passive treatments should not be used in isolation and they have a limited role in recurrent and persistent LBP disorders (see below). Passive therapies include the following.

Pharmacological treatment

There is some evidence that NSAIDs are effective for short-term symptomatic relief in patients with LBP.¹⁴⁰ There is no evidence that one type of NSAID is more effective than another. There is no evidence to support the long-term use of NSAIDs, both in relation to their lack of effectiveness

and their significant incidence of side effects. Other medications with good evidence of short-term effectiveness for pain control include skeletal muscle relaxants for acute LBP and tricyclic antidepressants for chronic LBP.¹⁴¹ Gabapentin has fair evidence for effective pain relief for radicular type pain.¹⁴¹

Mobilisation and manipulation

Mobilisation and manipulation techniques may be helpful for short-term pain relief (Chapter 17) and to restore spinal movement impairments detected on examination. These techniques affect multiple structures and can be adapted to the pattern of movement impairment that presents. Mobilisation techniques used in the treatment of LBP include:

- posteroanterior (PA) central pressure (Fig. 29.15a)
- posteroanterior unilateral pressure (Fig. 29.15b)
- rotational mobilisation (Fig. 29.15c).

Posteroanterior central mobilisation and rotations are used in patients with central or bilateral pain with movement impairments into extension. Posteroanterior unilateral mobilisation and rotation mobilisation techniques may be

continued

used for unilateral pain with sagittal and/or frontal plane movement impairments. The strength of the mobilisation technique used will depend on the level of tissue sensitivity and response to treatment. Mobilisation techniques can also be used to reassure the patient that movement is beneficial and not to be feared.

Soft tissue treatments

Tension and hypersensitivity of the myofascial structures is common with LBP and are related to pain-related muscle guarding and associated tissue hyperalgesia. This commonly involves the back muscles (lumbar multifidus and erector spinae), quadratus lumborum, abdominal wall and hip muscles (flexors, abductors and extensors). Tender focal areas of tension/'thickening' in the myofascial structures may be palpated.

Treatment of these areas with the use of soft tissue techniques and dry needling can result in short-term pain reduction and relaxation of guarded muscles. Soft tissue techniques may consist of transverse gliding, sustained longitudinal pressure on the tense bands emanating away from the most sensitive point and sustained firm pressure on the painful point (Fig. 29.16). Treatment is aimed at reducing tissue tension and sensitivity.



(a)



(b)



(c)

Figure 29.15 Mobilisation techniques (a) Posteroanterior central—The therapist performs an oscillating movement over the spinous processes using thumbs or heels of the hands. Elbows are extended and pressure is exerted through the shoulders and arms (b) Posteroanterior unilateral—Thumbs are placed over the apophyseal joints (c) Manipulation—rotation. With the patient positioned as shown, the therapist exerts a short sudden forward thrust on the ilium while maintaining strong counterpressure on the shoulder. This position can be used for manipulation or mobilisation



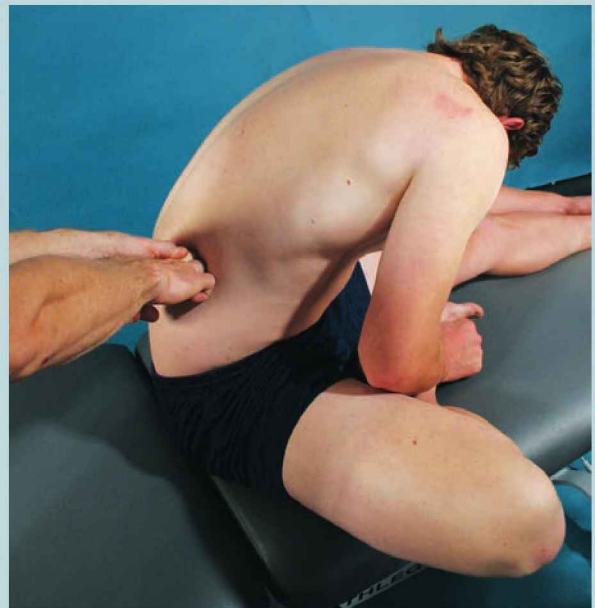
(a)

Dry needling to sensitised myofascial structures (Fig. 29.17) may also reduce pain and muscle tension in the short term, thus providing a window to facilitate the restoration of movement and function.

After treatment directed to either the articular and/or myofascial structures, the patient should be provided mobility exercises to maintain and restore the functional capacity.

Neurodynamic treatments

Neural tissue sensitisation may be linked to an LBP disorder when it is associated with peripheral referral of pain into the leg. If neural provocation tests such as the slump test and straight leg raise are limited (reflecting protective motor responses to a sensitised nerve), aggravate the patient's peripheral symptoms, and nerve palpation is associated with hyperalgesic responses, this may indicate peripheral sensitisation of the epineurial structures. Management of these pain states may respond to mobilisation of the articular and myofascial structures proximal to the nerve tissue (see above). Gentle mobilisation of the neural tissue (Fig. 29.18), if the structures are not highly sensitised, may also be



(b)

Figure 29.16 Soft tissue techniques (a) Sustained ischaemic pressure at each segmental level (b) Sustained ischaemic pressure using the knuckles to the quadratus lumborum in the position of sustained stretch

beneficial. However, 'stretching' of the nerve is **not** advisable as it may result in provocation of the disorder

Spinal injections: What is the evidence?

Where localised mechanically provoked pain persists, acting as a barrier for functional restoration, spinal injections may create a window of relief to enhance functional capacity. It must be noted that these techniques are usually short acting and so must be integrated with cognitive functional restoration programs. Some clinicians argue that this should occur once the pain has become chronic (i.e. 3-month duration) while others argue that interventional techniques may aid in the identification of the pain generator and may help expedite the athlete's recovery.¹³⁵

When compared to target-specific diagnoses achieved through accurate local anaesthetic injections done under image intensifier, our ability to make a specific diagnosis using clinical skill or imaging is very limited.

Structures that may become sensitised with LBP are:

- zygapophyseal joints
- intervertebral discs

continued



Figure 29.17 Dry needling of trigger points in the paraspinal and gluteal muscles

- sacroiliac joints
- nerve root sensitisation (i.e. pain radiating into the lower extremity without neurological signs).

These areas may be diagnosed and at times treated by the use of specific interventions outlined below.

Zygapophyseal joint pain can be diagnosed with medial branch blocks as the medial branches of the dorsal rami of the spinal nerves above and below the joint innervate each lumbar zygapophyseal joint. These procedures involve using an image intensifier to place a needle onto the appropriate medial branches that supply a zygapophyseal joint and injecting local anaesthetic onto the nerve. To block one joint, two nerves must be blocked; to block two adjacent joints, three nerves must be blocked. The patient then completes a pain chart and if the pain goes away for the duration of the anaesthetic, the clinician can be approximately 60% sure of the diagnosis. A second block with a different local anaesthetic is then done. If there is concordant pain relief, then the diagnostic confidence is approaching 90%.¹⁴²



Figure 29.18 Gentle mobilisation of the neural tissue

If the two medial branch blocks confirm the pain generator is the zygapophyseal joint, a radiofrequency neurotomy procedure can be performed. Radiofrequency neurotomy utilises an electrical current to ablate the medial branches to provide long-term symptom relief from zygapophyseal joint mediated pain.¹³⁵ This technique has a 90% chance of 60% pain relief and a 60% chance of 90% pain relief for lumbar zygapophyseal pain.¹⁴³ Note that zygapophyseal joint injections lack diagnostic and therapeutic validity.

Diagnostic criteria for *discogenic pain* are not clearly defined.¹³⁵ While discography is available and often used to aid in the diagnosis of discogenic pain, it lacks diagnostic validity and reliability.¹⁴⁴ Additionally, clinicians should be aware that discography can lead to accelerated progression of degenerative changes and carries a significant risk of infection.^{145, 146} Several interventional options have been proposed to treat discogenic pain including nucleoplasty, corticosteroid injection, anaesthetic injection and intradiscal electrothermal therapy (IDET). However, there is little quality evidence to support these interventions.¹³⁵

Sacroiliac joint pain can be diagnosed with a sacroiliac joint injection of anaesthetic performed with image guidance. With a control block, diagnostic confidence can approach 90%. An injection of anaesthetic and corticosteroid into the sacroiliac joint can provide symptomatic relief. Additionally, radiofrequency neurotomy can be performed for sacroiliac joint mediated pain with evidence for a 60% chance of more than 50% pain relief and with a 30% chance of 90% pain relief.¹⁴⁷ While some people advocate prolotherapy for sacroiliac joint problems, evidence for this is currently inconclusive.

Selective *nerve root blocks* using only local anaesthetic can be helpful to confirm a particular nerve root as a source for radicular leg pain as well as provide pain relief as stated earlier (radicular pain with nerve compression section).

Recurrent/persistent low back pain: a clinical sub-grouping approach

For a significant group of people (10–15%), LBP may persist beyond natural healing time following a tissue injury. For others, LBP develops insidiously and becomes persistent. This group of people together consume the majority of the healthcare resources.

PRACTICE PEARL

Genetic, physical (maladaptive movement patterns), neurophysiological (tissue sensitisation), psychosocial (fear, catastrophising, low self-efficacy, poor coping responses, stress, anxiety and depression) and lifestyle (sleep deficits and disruption, fatigue) are all risk factors for LBP persistence.

A combination of a person's negative thoughts and emotional and behavioural responses to pain (such as protective muscle guarding and/or avoidance) can set up a vicious cycle of pain sensitivity and disability.

Screening and identification of these factors early in the presentation of an LBP disorder is important to target treatment and prevent chronicity. Passive therapies for symptom palliation in this group should be avoided.

PRACTICE PEARL

Persistent LBP disorders can be broadly sub-grouped into low-, medium- and high-complexity categories (Fig. 29.2).^{21, 148}

In the rest of this chapter we outline the approach to this common group of patients according to the clinical subgroupings shown in Figure 29.2.

Low-complexity profile

A large group of athletes will present with low-complexity profiles. This profile is characterised by low to moderate levels of pain and disability, and with minimum distress. They predominantly have localised LBP with proportionate pain responses linked to specific movements, postures and activities (i.e. pain is provoked with spinal movements and loading but relieved when movements are ceased). They are commonly able to continue to participate with sport and have high levels of self-efficacy and active pain coping strategies. They present as low risk on psychosocial screening questionnaires.

Factors underlying this disorder may be ongoing sensitisation of spinal structures due to sport demands and training loads coupled with intrinsic physical factors (provocative motor control patterns +/- deconditioning). This may also be coupled with other sensitising factors

such as sleep disturbance or deficits and fatigue. Identification of the underlying modifiable drivers of the disorder is important for management.

Clinical presentation

Athletes with LBP complain of pain that is usually intermittent (provoked and relieved with specific postures, movements and activities), or constant only when following sustained provocative activities. Pain is relieved quickly when the provocative movement or activity ceases. The pain may be central or unilateral and is often described by the patient as a 'band across the lower back' +/- somatic pain in the buttock and/or hamstring.

On examination, impairments in spinal movement are uncommon; however, motor control impairments (patterns of movement linked to end-range strain of the lumbar spine) are usually observed. These clinical presentations are characterised by clear directional patterns of pain provocation.^{12, 149} They can be identified by trained therapists and present an opportunity for targeted rehabilitation.^{150, 151}

The following patterns are to illustrate the heterogeneous nature of lumbar spine motor control. While they are common clinical patterns, the clinician should not rigidly adhere to these presentations, but rather use them as a broad framework to guide management.

Active extension pain pattern

Some people with LBP present with a pattern of pain sensitivity-associated extension loading and movement activities associated with maintaining lumbar hyperlordosis with co-contraction of the back extensors and abdominal wall muscles. This is called an 'active extension pattern' where a person actively holds their back in lordosis during functional tasks (such as sitting, forward and backward bending and lifting), resulting in extension loading of sensitised spinal structures (Fig. 29.19). This is also often associated with an inability to relax the back muscles during forward bending in standing and sitting.¹³

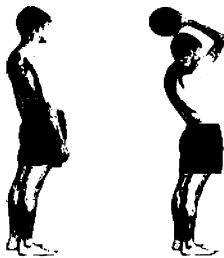
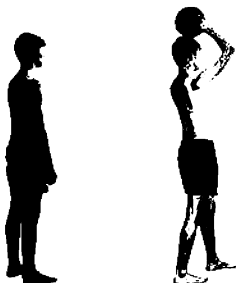
These pain disorders may be exacerbated with loaded squat exercises in the gym, and sports that involve spine extension coupled with side bending and extension such as tennis, fast bowling in cricket, throwing sports, swimming, running, dance and gymnastics. These patterns may be associated with conditioning deficits of the posterior hip muscles (gluteal and hamstrings) as well as tension of the hip flexors, resulting in anterior pelvic rotation. Spinal repositioning deficits in these cases demonstrate a tendency to reposition into lumbar extension from neutral.

On palpation, these presentations are commonly associated with increased muscle tension of the back muscles and quadratus lumborum and a lack of flexibility of the hip flexors. They report relief with lumbar flexion postures and movements.

(a) Active extension pattern-pain provoked with lumbar extension movements and loading**(b) Active extension pattern-corrected**

Figure 29.19 Active extension pattern (a) An 'active extension pattern' where a person actively holds their back in lordosis during functional tasks (such as sitting, forward and backward bending and lifting), resulting in extension loading of sensitised spinal structures (b) Correcting the active extension pattern with focus on posterior rotation of the pelvis and relaxation of the back extensor muscles, in order to reduce extension loading

(a) Passive extension pattern
- pain provoked with extension movements and loading

**(b) Passive extension pattern-corrected****Passive extension pain pattern**

The other presentation related to extension pain sensitivity is associated with a sway posture in standing and a tendency to hinge through the lower lumbar spine, while maintaining the thoracic spine in flexion, during backward bending (Fig. 29.20). This results in focal stress to the lower lumbar spine and is commonly associated with the sports that involve backward bending, especially throwing sports, ballet and gymnastics and/or sports that involve overhead arm activities (such as tennis and throwing sports).

These cases often present with conditioning deficits of the iliopsoas and thoracic erector spinae muscles, and poor eccentric control of the upper abdominal wall muscles (tendency to maintain thoracic flexion). Spinal repositioning deficits in these cases demonstrate a tendency to reposition into thoracic flexion.

On palpation, these presentations are commonly associated with increased muscle tension of the upper

Figure 29.20 Passive extension pattern (a) Sway posture in standing and a tendency to hinge through the lower lumbar spine, while maintaining the thoracic spine in flexion, during backward bending (b) Correcting their sway posture with thoracic extension and upper abdominal wall relaxation, during standing and backward bending

abdominal wall muscles. They report relief with lumbar flexion and when correcting their sway posture with thoracic extension and upper abdominal wall relaxation, during standing and backward bending (i.e. to reduce lower lumbar extension) (Fig. 29.20).

Flexion pain pattern

In other cases, where LBP has a flexion pattern of pain sensitivity, this is associated with a tendency to initiate forward bending and lifting from the lower back while fixing the pelvis into posterior rotation. This may either be associated with thoracolumbar flexion or lower lumbar flexion with thoracolumbar extension and upper abdominal wall activation (Fig. 29.21).^{12, 13, 152} This results in flexion loading stress to the lumbar spine and is associated with activities such as loaded squats in the gym and sports such as cycling, rowing and field hockey.

These cases often present with conditioning deficits of the hip flexors and lumbar back extensors and over-activation of the trunk flexors during bending activities. Spinal repositioning deficits in these cases demonstrate a tendency to reposition into lumbar flexion from neutral. On palpation, these presentations are commonly associated with increased tension, loss of eccentric relaxation and/or loss of flexibility of the gluteal and hamstring muscles.

Lateral pain pattern

In other cases, LBP is associated with pain sensitivity related to lateral loading and spinal movements in the frontal plane. These presentations are associated with frontal

plane motor control deficits of the hip (Trendelenburg) and/or thoracolumbar (lateral shift), which results in lateral loading strain to the low back (Fig. 29.22). These cases often present with conditioning deficits of the hip abductor (gluteal) muscles on the side of pain. Spinal repositioning deficits in these cases demonstrate a tendency to reposition into a lateral shifted position.

On palpation, these presentations are commonly associated with increased tension of the quadratus and lateral abdominal wall muscles on the side of pain. These presentations are common in sports related to limb loading such as running and jumping, landing sports.

Rotational plane pain pattern

In other cases, LBP is associated with pain sensitivity related to rotational movement and activities. This may be seen in sports such as tennis, golf and hockey. In many cases, these pain disorders are linked to control deficits in other planes (i.e. active extension, flexion, frontal plane). Lumbar spine rotation is more limited when the spine is at the end range of flexion, extension or side bending, and therefore the addition of rotation to end-range sagittal postures may increase spinal loading to the lumbar spine (Fig. 29.23).

In other cases, rotational strain may be placed on the spine due to a dynamic impairment of hip or thoracic spine rotation, resulting in increased rotational strain being placed on the lumbar spine. Assessment of the relative flexibility and control through the whole kinetic chain is important when assessing these disorders.

(a) Flexion pattern-pain provoked with lumbar flexion movements and loading



(b) Flexion pattern-corrected



Figure 29.21 Flexion pattern (a) Lower lumbar flexion with thoracolumbar extension and upper abdominal wall activation resulting in flexion loading (b) Correction of the flexion pattern via anterior pelvic tilt and relaxation of the thoracic spine into more flexed posture

(a) Lateral control pattern-pain provoked with lateral lumbar movements and loading**(b) Lateral control pattern-corrected**

Figure 29.22 Lateral control pattern (a) Lateral pain pattern associated with frontal plane motor control deficits of the hip (Trendelenburg) and/or thoracolumbar (lateral shift), which results in lateral loading strain to the low back (b) Correction of the frontal plane control focuses on gluteal activation and correction of the Trendelenburg and/or lateral shift



Figure 29.23 The addition of rotation to end-range sagittal postures (e.g. extension) may increase lumbar spine loading (image exaggerated for illustrative purposes)

Multidirectional pain pattern

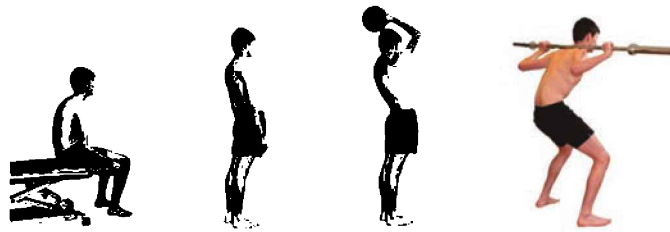
In other cases, LBP may present where the spine is sensitive to movement in more than one plane (i.e. flexion, extension, rotational or lateral). This may be associated with high levels of underlying tissue sensitivity and related to a combination of various motor control strategies as described previously (i.e. passive extension and flexion pattern). These people may have difficulty finding a pain-relieving movement direction (Fig. 29.24). Neutral spine positions with trunk muscle relaxation may provide some relief. Spinal repositioning deficits in these cases demonstrate a tendency to overshoot in different directions and these are often associated with high levels of thoracic muscle co-contraction (abdominal wall and back muscles).

Management

Management of these disorders involves targeting the cognitive, physical and lifestyle factors underlying the disorder.^{16, 138} The cognitive aspects of the intervention involve educating the athlete regarding the factors associated with LBP and providing reassurance regarding the spine's resilience. Manual therapy is only used if movement impairments are present and provide a barrier to functional activation.

Functional rehabilitation strategies are directed by the clinical presentation of the athlete as outlined below. Motor control training is integrated with the use of

- (a) **Multi-directional pattern (passive extension and flexion)**
- pain provoked with lumbar flexion and extension movements and loading



- (b) **Multi-directional-corrected**

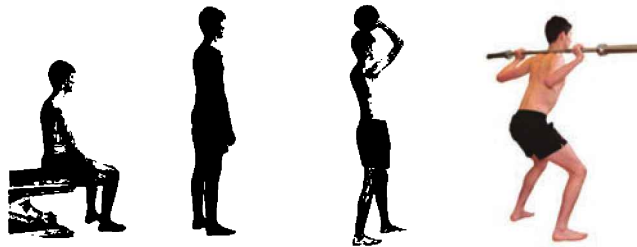


Figure 29.24 Multidirectional pattern (a) Pattern associated with pain provoked in more than one direction. In this case, a combination of both passive extension and flexion control patterns is observed (b) Correction of these patterns

mirrors and video so the athlete has a clear understanding of how they move. Postural correction should always reduce their LBP during provocative tasks.

The training is then integrated back into conditioning and sports-specific programs. Modifying training volume and technical aspects of the sport may be important to integrate with this. Lifestyle factors are also addressed when present.

Active extension pain pattern

The athlete is provided strategies to reduce their lumbar lordosis with posterior pelvic rotation exercises (initiated via the posterior hip muscles and not the abdominal wall). This may first be taught in supine crook lying and progressed to sitting to reduce anterior tilt and relax their back muscles. This is then progressed into sit-to-stand and squat positions (Fig. 29.19). In standing, posterior pelvic tilt is instructed in order to reduce the lumbar lordosis. If the athlete is unable to relax their back and/or hip muscles, then soft tissue work (Fig. 29.25a) and muscle stretching can be carried out (Fig. 29.25b).

These patterns are then integrated into conditioning-based exercises (squat and single-leg squat exercise) with weights that target the gluteal muscles, as well as sports-specific exercises (training forward and backward bending) (Fig. 29.19).

Passive extension pain pattern

The athlete is provided strategies to correct their postural sway in standing by increasing their thoracic

and upper lumbar lordosis with control of the pelvis augmented with the posterior hip muscles. If the athlete is unable to relax their upper abdominal wall and extend their thoracic spine, then soft tissue work directed to the upper abdominal wall can be used, and lateral costal diaphragm breathing exercise integrated with thoracic spine extension mobilisation techniques (Fig. 29.20).

Squatting is trained with thoracic extension and neutral lordosis. Backward bending is trained by facilitating upper lumbar and thoracic extension with hip extension in order to reduce focal extension stress. These movement patterns are then integrated with conditioning exercises that target postural control in a sport-specific exercise, with mirror feedback as appropriate.

Flexion pain pattern

The athlete is provided strategies to reduce lumbar flexion with anterior pelvic rotation exercises. This may first be taught in supine crook lying and progressed to sitting in order to facilitate anterior tilt without over-activation of the thoracic erector spinae and abdominal wall. This is then progressed into sit-to-stand and squat positions facilitating the movement with anterior pelvic rotation and a relaxed thoracic spine (Fig. 29.21).

These movement patterns are then integrated into conditioning-based exercises that reinforce these motor patterns (Fig. 29.21), as well as sports-specific exercises.



(a)



(b)

Figure 29.25 (a) Soft tissue therapy to lumbar extensors (b) Hip flexor stretch for active extension pattern

Utilising this cognitive functional approach has been found to be effective in rowers with flexion-related LBP.¹⁶

Lateral pain pattern

The athlete is provided strategies to control their frontal plane postural control in sitting, sit to stand and standing by centring their head position in relation to the pelvis. This is progressed to single-leg standing (with or without support), single-leg squatting with careful attention to control of the hip/pelvis (preventing Trendelenburg pattern) as well as the thoracolumbar spine alignment relative to the pelvis (nose in line with navel). Training can be integrated with conditioning exercises that target single-leg loading and load transfer, progressing towards dynamic loading and sports-specific exercises (Fig. 29.22).

Rotational plane pain pattern

The athlete is provided strategies to control their spine during provocative rotational movements. In some cases, this may involve adjustments to spinal posture as outlined earlier (Fig. 29.23). In other cases, this involves addressing impairments of movement of thoracic rotation (usually linked to co-contraction of the oblique abdominal wall muscles). This involves teaching relaxed thoracic spine rotation with breathing control (lateral costal breathing) during rolling, sitting, standing and functional rotational tasks. Where there is an impairment of hip rotation, addressing this with soft tissue work stretching and then active functional integration is trained. This can then be integrated into conditioning-based exercises with the use of pulleys.

Multidirectional pain pattern

The nature of the training is an integration of any of the previously mentioned motor control strategies dependent on the athlete's presentation (Fig. 29.24). For example, this may involve training the control of both the forward and backward bending movement and single-leg loading (see Fig. 29.2, central pillar). This is integrated into functional and conditioning programs as required.

Medium-complexity profile

A moderate number of athletes will present with medium-complexity profiles. (See Fig. 29.2, central pillar.)

Clinical presentation

This sub-group profile is characterised by moderate levels of pain, disability and distress. These athletes often avoid sporting activities due to pain. They may have a mixed pain picture with both pain at rest and with mechanical loading of the spine. Negative cognitive factors are dominant in these disorders such as negative beliefs (i.e. 'my discs are crumbling'), high levels of fear (i.e. 'I am too frightened to dance'), catastrophising (i.e. 'I think my back will collapse', 'it will never get better'), hypervigilance (i.e. 'I can't stop thinking about my

back pain'), low levels of self-efficacy (i.e. 'I don't have any way to control the pain', 'I can't play sport with the pain'). They may also report high levels of contextual stress (i.e. 'my current stress levels are 7-8/10').

These cognitive factors are related to pain amplification,¹⁵³ and in some may be linked to high levels of protective muscle guarding, slow cautious movement patterns and protective movement behaviours (i.e. bracing of the hands on the thigh with bending, breath holding, avoidance of limb loading). These movement behaviours are usually superimposed upon the movement patterns described under the low-complexity profile. With high levels of fear, it is common to see movement impairments (loss of functional range of movement), reflected in high levels of trunk muscle guarding associated with feared movement direction (Fig. 29.26).

In other cases, the level of distress may be disproportionate to the clinical presentation (i.e. high levels of pain and distress, but with pain-free movement and functional activity, no tissue hyperalgesia but pain with attention and at rest).¹⁵⁴ Lifestyle factors such as high levels of contextual stress, activity avoidance and poor sleep hygiene are common. These athletes present as medium risk on screening questionnaires.



(a)



(b)

Figure 29.26 Protective patterns (a) Protective pattern with thoracic extension and bracing of the hand (b) Protective pattern with lumbar extension and bracing of the hand

Management

Education regarding the multidimensional factors associated with LBP is essential for the person to understand that their LBP is not related to 'damage', but due to sensitised spinal structures related to negative thoughts, fears and pain-related behavioural responses. Reassurance regarding the spine's resilience is essential.

PRACTICE PEARL

High levels of fear and catastrophising are addressed through education as well as providing the athlete with pain-control strategies such as relaxation, mindfulness techniques and graded movement training in a relaxed manner back towards fear activities where this is a barrier.

This is directed by their pain-related movement profile as outlined in the low-complexity group. (See also Fig. 29.2.)

Where movement impairments prevent functional activation, manual therapy and soft tissue techniques can be used to relax guarded muscles, so as to reinforce that spinal movement can be trusted. The athlete is gradually exposed to the feared movement pattern (i.e. fear of lumbar flexion is initially trained in supine with posterior pelvic tilt, hip flexion, lumbar flexion in stand and then progressed to standing). Once motor control patterns have been normalised, these are gradually integrated back into conditioning programs and then into the person's sport.

Where there is an absence of clinical signs linked to the patient's LBP, it is critical to reassure the athlete to reduce their focus on the back and trust the resilience of the spine. Manual therapies and specific exercises that direct attention to the back in these cases may be unhelpful.

Addressing lifestyle factors such as stress management (meditation techniques), sleep hygiene, alcohol intake and minimising sedentary behaviours is also important where indicated. Graduated physical activation is also central to this process where this has been avoided, and ideally this should be linked to their sport. This may require a graduated, paced process to recovery to avoid a boom-bust activity pattern.

High-complexity profile

A small group of athletes present with high-complexity profiles. (See Fig. 29.2, right column.)

Clinical presentation

This profile is characterised by moderate to high levels of pain, disability and distress. They will almost certainly be avoiding sporting and social activities due to their pain, or be exercising obsessively with pain. They usually have a mixed pain picture and may present as widespread and

constant pain that is exacerbated with minimal movement provocation. Consequently, they may easily flare up with minimal physical examination.

The other presentation as outlined earlier is the complaint that pain is distressing, but with an absence of clinical findings. In this profile pain is often worse at rest and when focused on, and better when moving and distracted. This profile is often linked to underlying anxiety and hypervigilance. These athletes also commonly report comorbid pains (headaches, migraine, stomach pain) as well as nonspecific health complaints (i.e. fatigue, lack of energy).

These athletes present with similar negative cognitive factors as outlined in the medium-complexity profile, (i.e. negative beliefs, high levels of fear, stress, hypervigilance, low self-efficacy and poor coping). They commonly have catastrophic beliefs about their pain (i.e. 'my pelvis is out', 'my disc is stuffed') coupled with high levels of perceptual body distortion. In addition, these athletes present with high levels of emotional distress, heightened anxiety levels and depressed mood. It is important to screen as to whether the athlete has a diagnosed comorbid anxiety and/or depressive disorder and if so whether this is being actively managed. These presentations usually present with maladaptive functional behaviours, muscle guarding and avoidance as outlined in the medium-complexity profile. This may also be coupled with hyperarousal (apical rapid breathing, agitation, sweating, fidgeting).

The athlete may also present with pain behaviours such as constant touching of the back, habitual self-manipulation and rubbing of the area, grimacing and an inability to relax. This may also be coupled with negative lifestyle factors (sleep disturbance, inactivity, alcohol intake and social isolation). These athletes may also tend to become reliant on passive therapies. They present as high risk on screening questionnaires.

Management

A cognitive functional approach that provides a biopsychosocial understanding of pain, pain control strategies through body relaxation, mindfulness, movement retraining and functional activation is central to this process.¹⁵⁵ Education regarding the multidimensional factors associated with LBP is essential so that the person understands that their thoughts, emotions and behaviours can influence their pain responses and behaviours. Reassurance regarding the spine's resilience is central to the therapeutic process. Before movement training can be initiated, the athlete may need to be trained to relax using slow relaxed-diaphragm breathing, body relaxation and mindfulness techniques.

The approach to movement training is then guided by their movement profile (outlined in low-complexity profile). Focus is placed on discouraging protective movement behaviours (propping of the hands, breath

holding and tensing with movement). Pain behaviours (i.e. self-manipulation, repeated stretching and touching of the spine) are also actively discouraged.

PRACTICE PEARL

Graduated physical activation and addressing maladaptive behaviours is usually more difficult in this group due to the dominance of central nervous system pain features. (See also Chapter 5.)

Consequently, minimal movement may result in pain flares and over-zealous clinicians may find the athlete distressed with a pain flare. Slow graduated physical activation with a clear paced program to return to sporting activity is usually required. Addressing lifestyle factors such as sleep disturbance, contextual life stress and activity avoidance is central to this process. The use of exercise and sleep diaries may assist this process.

The presence of uncontrolled comorbid anxiety, panic disorder, depression and/or high levels of contextual stresses in the athlete's social life may necessitate psychological referral. Central nervous system pain features and sleep disturbances may require medical referral for targeted pharmacological treatments to provide a window of opportunity for functional restoration.

Reliance on passive therapies and interventional medicine is usually unhelpful and provocative and should be avoided in these cases.

REHABILITATION FOLLOWING LOW BACK PAIN

Developing an ongoing plan based on the risk profile of the athlete is important in order to minimise the future risks. This includes managing *physical risk factors* such as training loads, technical skills, optimising body posture and control, conditioning and fitness specific to the sport's requirements. The nature of this training is specific to the individual presentation of the athlete. Integrating this into gym-based conditioning programs is crucial.

It also involves managing *psychosocial* and *lifestyle* risks such as energy levels, emotional and psychological health, stress resilience, incorporating relaxation strategies, sleep hygiene, healthy diet and minimising sedentary behaviours. There are many online resources to support both the clinician and the athlete in this process.

Sporting technique

Poor technique in sporting activities may increase stress on the structures of the lumbar spine. The technique should be assessed with the aid of a coach and in some cases with high-speed film and biomechanical analysis. Any necessary corrections should be made under supervision and with the

guidance of the coach. Examples of sport-specific technique factors linked to lower back pain are covered in Chapter 9.

Optimal motor control

There is a lot of controversy regarding core stability exercises in LBP. Systematic reviews support the idea that 'stabilisation' exercises are not more effective than general exercise for LBP.¹⁵⁶ Furthermore, studies investigating muscle changes suggest that changes in the stabilising muscles are not prognostic of superior outcomes. While trunk muscle strength and conditioning is critical for many sports, as previously reported, recurrent and chronic LBP is often associated with higher levels of co-contraction of the trunk muscles. Hence, rehabilitation programs need to be sports specific and attend to the individual impairments of the athlete rather than be prescriptive.

Control of the lumbar spine is influenced greatly by both the thoracic spine and the hip pelvic region. Hence, optimising control of the lower back involves attending to control of these proximal and distal regions. Adequate strength of the lower limbs is essential in sports involving limb loading to adequately dampen ground reaction forces and ensure trunk control. The double- and single-leg squat is an excellent rehabilitation exercise combining motor control and strengthening. Similarly, a lack of upper body (upper limb and thoracic) strength and control can compromise lumbar spine mechanics in sports demanding use of the upper limbs (e.g. tennis, cricket, throwing and swimming). In these cases, maintaining a balanced exercise program that ensures optimal control of these regions is important.

Flexibility

Depending on the sport and the individual, addressing the relative flexibility of the hip muscles and/or the thoracic/shoulder girdle regions, as well as the lumbar spine, is important. Commonly, tense or 'shortened' muscles (i.e. erector spinae, hip flexors, gluteals, hamstrings, rectus abdominis, quadratus lumborum, pectorals and latissimus dorsi) may influence the biomechanics and motor control strategies of the lumbar spine. For example, tightness of the hamstrings may result in greater flexion strain of the lower back in a rower. Conversely, tight hip flexors may result in increased lumbar spine extension in a dancer or runner. On the other hand, tightness of the latissimus dorsi may result in excessive side bending of the lumbar spine in a fast bowler in cricket. The tension in these muscles should be addressed as part of the rehabilitation program. The various techniques are shown in Figure 29.27.

CONCLUSION

The clinical approach to the athlete with LBP requires a multidimensional perspective. The first is to triage the athlete in order to rule out serious and specific

pathoanatomical disorders. The second is to identify the dominance and presence of potential physical, psychosocial and lifestyle risk factors that may be influencing pain and disability profiles both in the presence and absence of specific pathology. The integration of screening questionnaires is critical in this process.

PRACTICE PEARL

Considering non-specific LBP as a feature of sensitisation of the spine structures, rather than 'damage to the spine', is critical to change the clinician's and the athlete's mindset to their problem.

In the acute phase of a disorder, providing pain control, restoration of function and then targeting the intervention based on the individual's biopsychosocial profile allows for targeting of care. When LBP is persistent, identification of underlying modifiable drivers of pain sensitisation and disability is critical. This is tailored to the complexity profile of the athlete. In high-complexity cases, multidisciplinary care may be required.

There is growing evidence that an individualised approach is more effective than traditional structural, biomechanical approaches to LBP. Encouraging the athlete to develop a positive view of the spine's resilience, enhancing optimal motor control and confidence in their spine, as well as guiding the patient to make healthy lifestyle choices, are goals of this process.

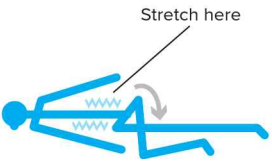
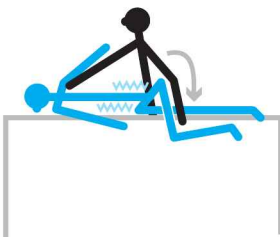
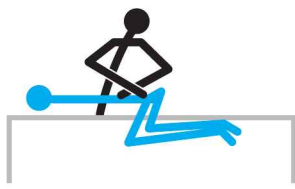
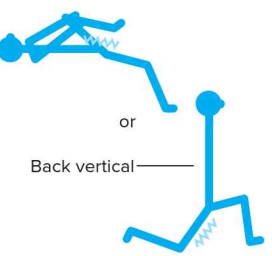
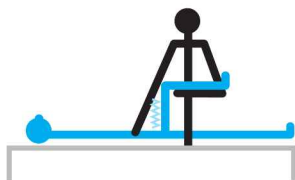
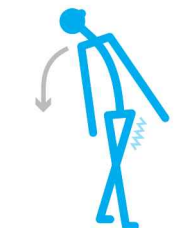
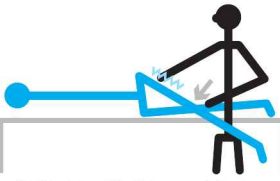
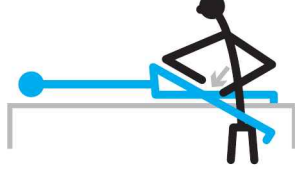
Muscle	Self-exercise	Assisted exercise	Myofascial release
Erector spinae			 Patient is side-lying. Therapist's wrists are crossed over each other to provide traction.
Psoas			 Therapist's hand is over the psoas. Hip is extended from the flexed position.
Iliotibial band		 Patient is side-lying and facing away. Hip is extended and adducted.	 Patient is side-lying and facing away. Therapist uses elbow/forearm to perform release.

Figure 29.27 Techniques used to treat tightness of individual muscles

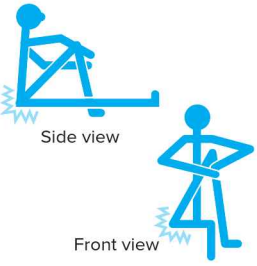
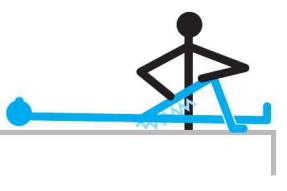
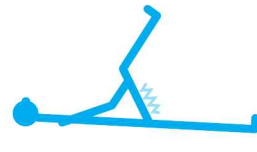
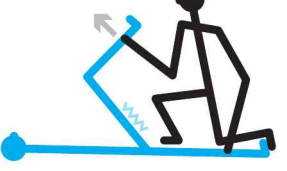
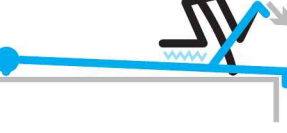
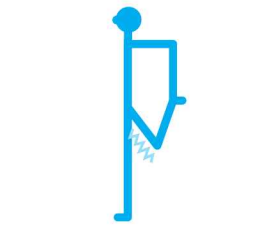
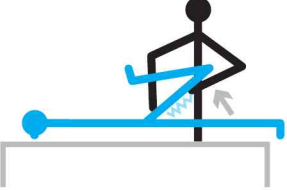
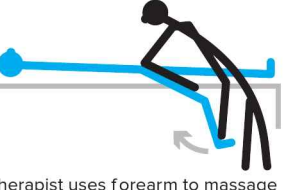
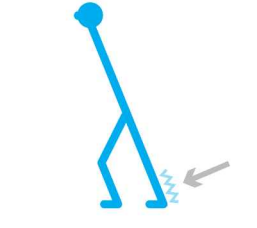
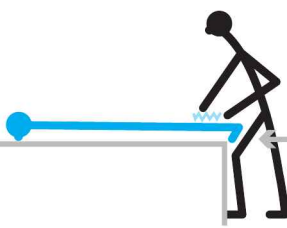
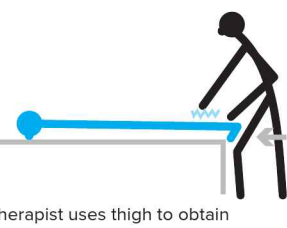
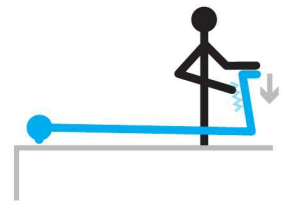
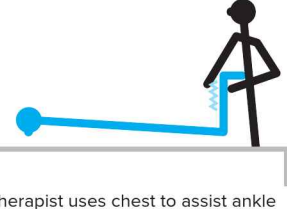
Muscle	Self-exercise	Assisted exercise	Myofascial release
Hip external rotators	 Side view Front view	 Hip into adduction with treatment leg crossed over opposite leg.	 Patient is side-lying. Therapist stands behind and takes the top leg backward.
Hamstrings			 Patient is prone. The elbow or forearm is kept stationary and the knee passively extended.
Rectus femoris		 Keep pelvis down while extending hip and flexing knee.	 Therapist uses forearm to massage up the thigh of the leg which is hanging off the table.
Gastrocnemius	 Pressure on back leg		 Therapist uses thigh to obtain passive ankle dorsiflexion.
Soleus	 Pressure on front leg		 Therapist uses chest to assist ankle dorsiflexion.

Figure 29.27 (cont.) Techniques used to treat tightness of individual muscles

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e